

Chapter 10 Part 3: K<sub>sp</sub>

1. A 10.0 g sample of  $\text{Pb}(\text{NO}_3)_2(s)$  is dissolved in distilled water and an excess of  $1.0\text{ M HCl}(aq)$  is added. The  $\text{PbCl}_2(s)$  formed is filtered and placed in an oven to dry. The final mass of the precipitate is significantly greater than predicted. Which of the following could account for the discrepancy?
- (A) The sample of  $\text{Pb}(\text{NO}_3)_2(s)$  was wet.
- (B) The  $K_{sp}$  of  $\text{PbCl}_2(s)$  is extremely small.
- (C) The precipitate was not dried to a constant mass.
- (D) A  $3.0\text{ M HCl}(aq)$  solution was used instead of a  $1.0\text{ M HCl}(aq)$  solution.

2.  $\text{BaCO}_3(s) \rightleftharpoons \text{Ba}^{2+}(aq) + \text{CO}_3^{2-}(aq) \quad K_{sp} = 8.1 \times 10^{-9}$

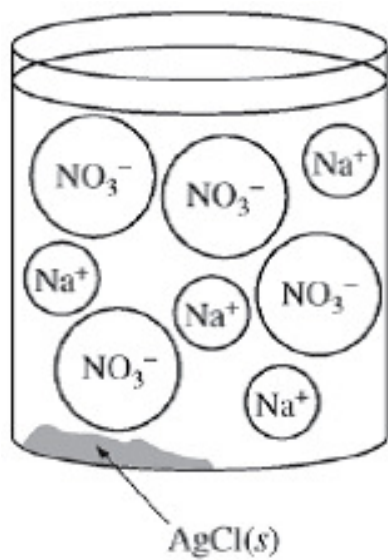
The dissolution of  $\text{BaCO}_3(s)$  in water is represented above. A 20.0 mL sample of  $1.0 \times 10^{-4}\text{ M Ba}(\text{NO}_3)_2(aq)$  is mixed with an 80.0 mL sample of  $1.0 \times 10^{-4}\text{ M Na}_2\text{CO}_3(aq)$ . Which of the following correctly predicts and explains the outcome?

- (A) A precipitate will form, because  $Q = 1.0 \times 10^{-8}$ .
- (B) A precipitate will form, because  $Q = 1.6 \times 10^{-9}$ .
- (C) A precipitate will not form, because  $Q = 1.0 \times 10^{-8}$ .
- (D) A precipitate will not form, because  $Q = 1.6 \times 10^{-9}$ .
3.  $\text{Ag}^+(aq) + \text{Cl}^-(aq) \rightleftharpoons \text{AgCl}(s)$

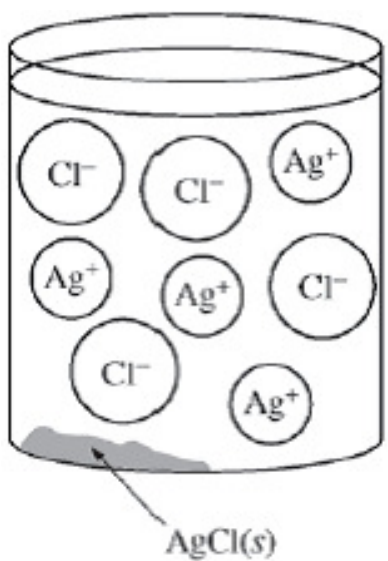
A student mixes dilute  $\text{AgNO}_3(aq)$  with excess  $\text{NaCl}(aq)$  to form  $\text{AgCl}(s)$ , as represented by the net ionic equation above. Which of the diagrams below best represents the ions that are present in significant concentrations in the solution? ( $K_{sp}$  for  $\text{AgCl}$  is  $1.8 \times 10^{-10}$ .)

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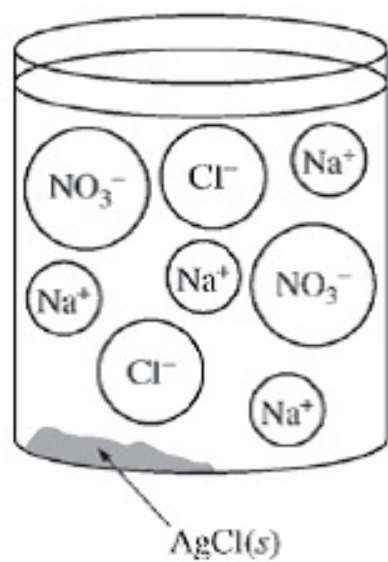
(A)



(B)

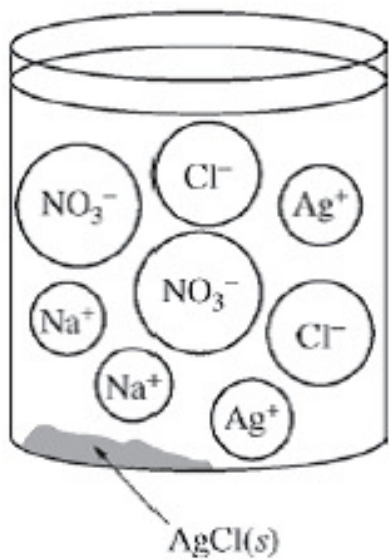


(C)



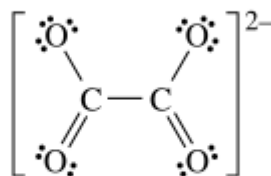
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(D)



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4. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



A Lewis electron-dot diagram of the oxalate ion,  $\text{C}_2\text{O}_4^{2-}$ , is shown.

- (a) Identify the hybridization of the valence orbitals of either carbon atom in the oxalate ion.
- (b) Silver oxalate,  $\text{Ag}_2\text{C}_2\text{O}_4(s)$ , is slightly soluble in water. The value of  $K_{sp}$  for  $\text{Ag}_2\text{C}_2\text{O}_4$  is  $5.40 \times 10^{-12}$ .
- (i) Write the expression for the solubility-product constant,  $K_{sp}$ , for  $\text{Ag}_2\text{C}_2\text{O}_4$ .
- (ii) Calculate the molar solubility of  $\text{Ag}_2\text{C}_2\text{O}_4$  in neutral distilled water.

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(iii) The molar solubility of  $\text{Ag}_2\text{C}_2\text{O}_4$  increases when it is dissolved in  $0.5M \text{HClO}_4(aq)$  instead of neutral distilled water. Write a balanced, net-ionic equation for the process that occurs between species in solution that contributes to the increased solubility of  $\text{Ag}_2\text{C}_2\text{O}_4(aq)$  in  $\text{HClO}_4(aq)$ .

5. What is the molar solubility in water of  $\text{Ag}_2\text{CrO}_4$ ? (The  $K_{sp}$  for  $\text{Ag}_2\text{CrO}_4$  is  $8 \times 10^{-12}$ .)
- (A)  $8 \times 10^{-12} M$   
(B)  $2 \times 10^{-12} M$   
(C)  $\sqrt{4 \times 10^{-12}} M$   
(D)  $\sqrt[3]{4 \times 10^{-12}} M$   
(E)  $\sqrt[3]{2 \times 10^{-12}} M$
6. Several reactions are carried out using  $\text{AgBr}$ , a cream-colored silver salt for which the value of the solubility-product constant,  $K_{sp}$ , is  $5.0 \times 10^{-13}$  at 298 K.
- Write the expression for the solubility-product constant,  $K_{sp}$ , of  $\text{AgBr}$ .
  - Calculate the value of  $[\text{Ag}^+]$  in 50.0 mL of a saturated solution of  $\text{AgBr}$  at 298 K.
  - A 50.0 mL sample of distilled water is added to the solution described in part (b), which is in a beaker with some solid  $\text{AgBr}$  at the bottom. The solution is stirred and equilibrium is reestablished. Some solid  $\text{AgBr}$  remains in the beaker. Is the value of  $[\text{Ag}^+]$  greater than, less than, or equal to the value you calculated in part (b)? Justify your answer.
  - Calculate the minimum volume of distilled water, in liters, necessary to completely dissolve a 5.0 g sample of  $\text{AgBr}(s)$  at 298 K. (The molar mass of  $\text{AgBr}$  is  $188 \text{ g mol}^{-1}$ .)
  - A student mixes 10.0 mL of  $1.5 \times 10^{-4} M \text{AgNO}_3$  with 2.0 mL of  $5.0 \times 10^{-4} M \text{NaBr}$  and stirs the resulting mixture. What will the student observe? Justify your answer with calculations.
  - The color of another salt of silver,  $\text{AgI}(s)$ , is yellow. A student adds a solution of  $\text{NaI}$  to a test tube containing a small amount of solid, cream-colored  $\text{AgBr}$ . After stirring the contents of the test tube, the student observes that the solid in the test tube changes color from cream to yellow.
    - Write the chemical equation for the reaction that occurred in the test tube.
    - Which salt has the greater value of  $K_{sp}$ :  $\text{AgBr}$  or  $\text{AgI}$ ? Justify your answer.

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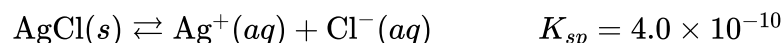
7. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of  $\text{Ag}^+(aq)$  in a solution of  $\text{AgNO}_3(aq)$ . The student adds an excess of  $8.0\text{ M HCl}(aq)$  to  $200.0\text{ mL}$  of the solution, causing  $\text{AgCl}(s)$  to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

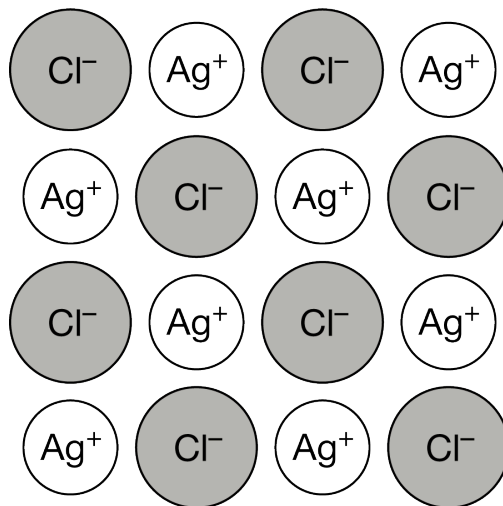
|                                                                  |         |
|------------------------------------------------------------------|---------|
| Initial mass of filter paper                                     | 0.851 g |
| Mass of filter paper + $\text{AgCl}(s)$ precipitate after drying | 2.311 g |

- (a) Calculate the number of moles of  $\text{AgCl}(s)$  that precipitated. (The molar mass of  $\text{AgCl}$  is  $143\text{ g/mol}$ ).
- (b) Assuming that all of the  $\text{Ag}^+(aq)$  ions precipitated, calculate the original molarity of  $\text{Ag}^+(aq)$  in the solution before  $\text{HCl}(aq)$  was added.
- (c) The calculated concentration of  $\text{Ag}^+(aq)$  from the student's data is higher than the actual concentration of the original solution. The student claims that the calculated concentration is too high because the original solution contained some  $\text{Na}^+(aq)$  impurities that resulted in some  $\text{NaCl}(s)$  precipitating along with the  $\text{AgCl}(s)$ . Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate,  $\text{AgCl}(s)$ , is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of  $\text{AgCl}(s)$  in water and a  $K_{sp}$  value at the temperature of the water.

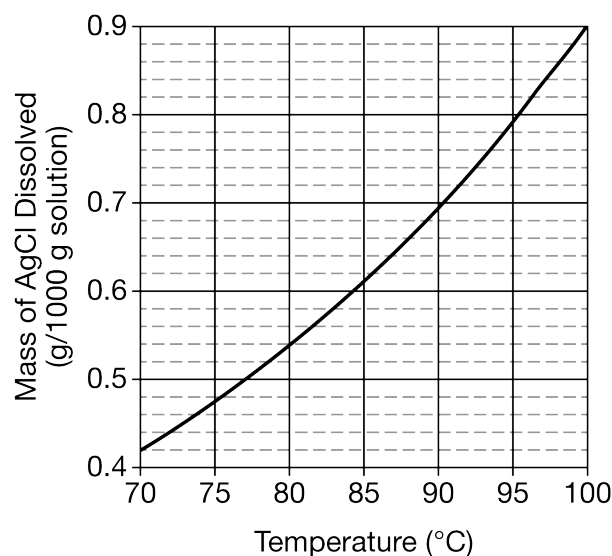


- (d) Calculate  $[\text{Ag}^+]$  in the student's solution.
- (e) The value of  $K_{sp}$  of another silver salt,  $\text{AgBr}(s)$ , is  $5.0 \times 10^{-13}$ . If separate, saturated solutions of  $\text{AgBr}$  and  $\text{AgCl}$  are prepared, which solution will have the greater  $[\text{Ag}^+]$ ? Justify your answer.
- (f) The student finds the following particulate model of  $\text{AgCl}(s)$ . Assuming the crystal structures are similar, how should the student modify the model to represent  $\text{AgBr}(s)$ ? Justify your answer in terms of the radius and arrangement of electrons in the ions.

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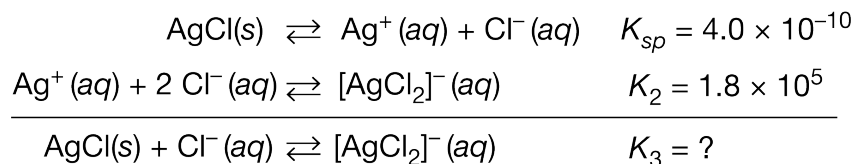
(g) The enthalpy of dissolution of  $\text{AgCl}(s)$  in water is a positive value. Based on this information, the student claims that the energy released when separating the ions in  $\text{AgCl}(s)$  is greater than the energy released during the hydration of  $\text{Ag}^+$  and  $\text{Cl}^-$  by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of  $\text{AgCl}$  in a  $2.5\text{ M NaCl}$  solution at various temperatures. Some  $\text{AgCl}(s)$  remains in the container at all temperatures studied. The data are plotted on the graph below.



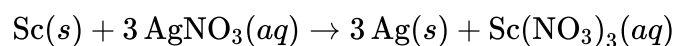
(h) A 1000. g sample of a saturated solution at  $95^\circ\text{C}$  is cooled to  $85^\circ\text{C}$ . Determine the mass of  $\text{AgCl}(s)$  that would precipitate out of the solution.

(i) The specific heat capacity of the solution is  $3.3\text{ J}/(\text{g} \cdot ^\circ\text{C})$ . How much heat is released when 1000. g of the solution is cooled from  $95^\circ\text{C}$  to  $85^\circ\text{C}$ ?

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(j) The student finds that AgCl is more soluble in NaCl(aq) than in water because of the series of reactions above. Determine the value of  $K_3$  for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic scandium, Sc(s), in a solution of AgNO<sub>3</sub>(aq) and observes that the following reaction occurs.



(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the Sc(s) decreases significantly as it sits in the AgNO<sub>3</sub> solution. The student claims that the mass decreases because scandium ions dissolve into the solution as the experiment progresses. Does the student's claim accurately explain the decrease in the mass of the Sc(s)? Justify your answer.



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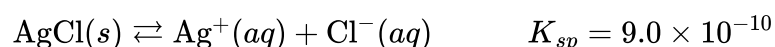
8. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of  $\text{Ag}^+(aq)$  in a solution of  $\text{AgNO}_3(aq)$ . The student adds an excess of  $2.0\text{ M HCl}(aq)$  to  $300.0\text{ mL}$  of the solution, causing  $\text{AgCl}(s)$  to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

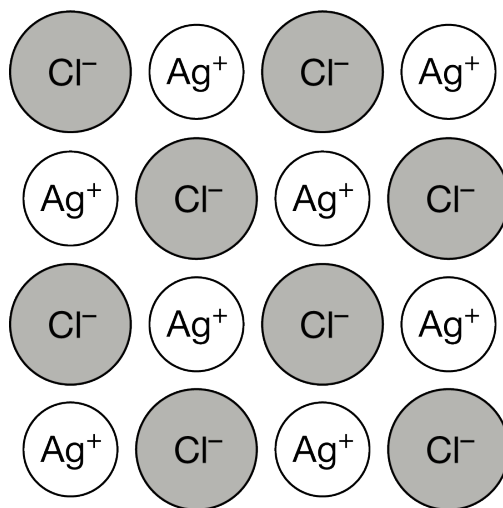
|                                                                  |         |
|------------------------------------------------------------------|---------|
| Initial mass of filter paper                                     | 0.650 g |
| Mass of filter paper + $\text{AgCl}(s)$ precipitate after drying | 0.793 g |

- (a) Calculate the number of moles of  $\text{AgCl}(s)$  that precipitated. (The molar mass of  $\text{AgCl}$  is  $143\text{ g/mol}$ ).
- (b) Assuming that all of the  $\text{Ag}^+(aq)$  ions precipitated, calculate the original molarity of  $\text{Ag}^+(aq)$  in the solution before  $\text{HCl}(aq)$  was added.
- (c) The calculated concentration of  $\text{Ag}^+(aq)$  from the student's data is lower than the actual concentration of the original solution. The student claims that the calculated concentration is too low because some of the  $\text{AgCl}(s)$  passed through the filter paper. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate,  $\text{AgCl}(s)$ , is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of  $\text{AgCl}(s)$  in water and a  $K_{sp}$  value at the temperature of the water.

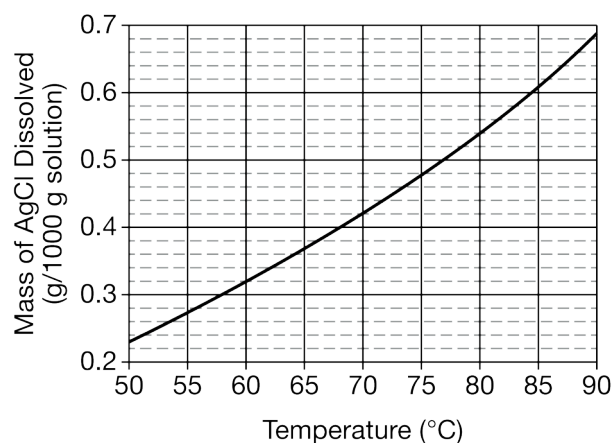


- (d) Calculate  $[\text{Ag}^+]$  in the student's solution.
- (e) The value of  $K_{sp}$  of another silver salt,  $\text{AgF}(s)$ , is  $2.0 \times 10^2$ . If separate, saturated solutions of  $\text{AgF}(s)$  and  $\text{AgCl}(s)$  are prepared, which solution will have the greater  $[\text{Ag}^+]$ ? Justify your answer.
- (f) The student finds the following particulate model of  $\text{AgCl}(s)$ . Assuming the crystal structures are similar, how should the student modify the model to represent  $\text{AgF}(s)$ ? Justify your answer in terms of the radius and arrangement of electrons in the ions.

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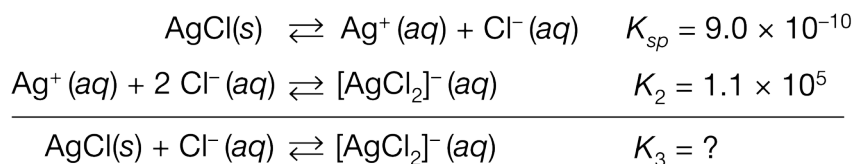
(g) The enthalpy of dissolution of  $\text{AgCl}(s)$  in water is a positive value. Based on this information, the student claims that the energy required to separate the ions in  $\text{AgCl}(s)$  is equal to the energy released during the hydration of  $\text{Ag}^+$  and  $\text{Cl}^-$  by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of  $\text{AgCl}$  in a  $2.5\text{ M}$   $\text{NaCl}$  solution at various temperatures. Some  $\text{AgCl}(s)$  remains in the container at all temperatures studied. The data are plotted on the graph below.



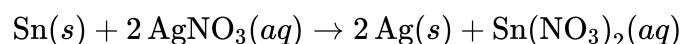
(h) A 1000. g sample of a saturated solution at  $85^\circ\text{C}$  is cooled to  $50.^\circ\text{C}$ . Determine the mass of  $\text{AgCl}(s)$  that would precipitate out of the solution.

(i) The specific heat capacity of the solution is  $3.4\text{ J}/(\text{g} \cdot ^\circ\text{C})$ . How much heat is released when 1000. g of the solution is cooled from  $85^\circ\text{C}$  to  $50.^\circ\text{C}$ ?

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(j) The student finds that AgCl is more soluble in NaCl(aq) than in water because of the series of reactions above. Determine the value of  $K_3$  for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic tin, Sn(s), in a solution of AgNO<sub>3</sub>(aq) and observes that the following reaction occurs.



(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the Sn(s) decreases significantly as it sits in the AgNO<sub>3</sub> solution. The student claims that the mass decreases because the atoms in Sn(s) are converted to atoms of Ag(s) as the experiment progresses, and Ag(s) has a smaller molar mass than Sn(s) has. Does the student's claim accurately explain the decrease in the mass of the Sn(s)? Justify your answer.

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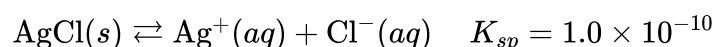
9. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of  $\text{Ag}^+(aq)$  in a solution of  $\text{AgNO}_3(aq)$ . The student adds an excess of  $3.0\text{ M HCl}(aq)$  to  $100.0\text{ mL}$  of the solution, causing  $\text{AgCl}(s)$  to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

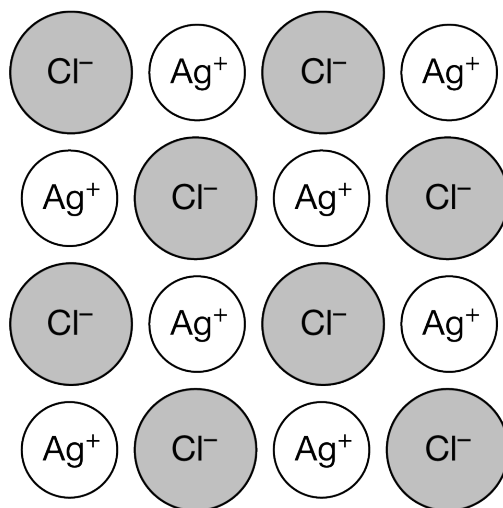
|                                                                  |         |
|------------------------------------------------------------------|---------|
| Initial mass of filter paper                                     | 0.718 g |
| Mass of filter paper + $\text{AgCl}(s)$ precipitate after drying | 3.578 g |

- (a) Calculate the number of moles of  $\text{AgCl}(s)$  that precipitated. (The molar mass of  $\text{AgCl}$  is  $143\text{ g/mol}$ ).
- (b) Assuming that all of the  $\text{Ag}^+(aq)$  ions precipitated, calculate the original molarity of  $\text{Ag}^+(aq)$  in the solution before  $\text{HCl}(aq)$  was added.
- (c) The calculated concentration of  $\text{Ag}^+(aq)$  from the student's data is lower than the actual concentration of the original solution. The student claims that the calculated concentration is too low because the filter paper and  $\text{AgCl}(s)$  precipitate were not fully dry. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate,  $\text{AgCl}(s)$ , is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with cold, distilled water to make a saturated solution. The student finds the following equation for the dissolution of  $\text{AgCl}(s)$  in water and a  $K_{sp}$  value at the temperature of the water.

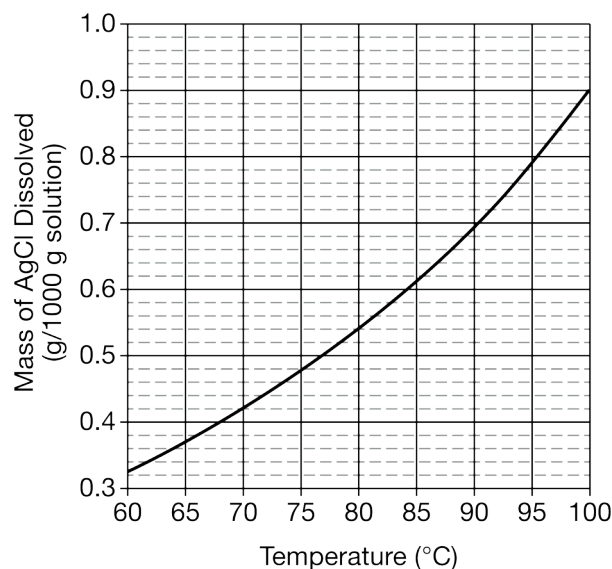


- (d) Calculate  $[\text{Ag}^+]$  in the student's solution.
- (e) The value of  $K_{sp}$  of another silver salt,  $\text{AgI}(s)$ , is  $8.3 \times 10^{-17}$ . If separate, saturated solutions of  $\text{AgI}$  and  $\text{AgCl}$  are prepared, which solution will have the greater  $[\text{Ag}^+]$ ? Justify your answer.
- (f) The student finds the following particulate model of  $\text{AgCl}(s)$ . Assuming the crystal structures are similar, how should the student modify the model to represent  $\text{AgI}(s)$ ? Justify your answer in terms of the radius and arrangement of electrons in the ions.

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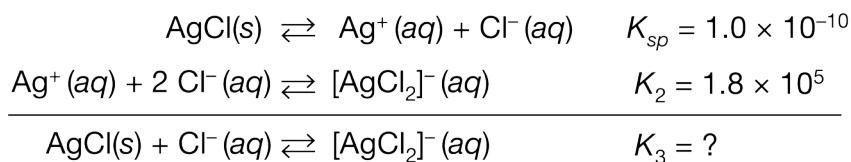
(g) The enthalpy of dissolution of  $\text{AgCl}(s)$  in water is a positive value. Based on this information, the student claims that the energy required to separate the ions in  $\text{AgCl}(s)$  is greater than the energy released during the hydration of  $\text{Ag}^+$  and  $\text{Cl}^-$  by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of  $\text{AgCl}$  in a  $2.5\text{ M NaCl}$  solution at various temperatures. Some  $\text{AgCl}(s)$  remains in the container at all temperatures studied. The data are plotted on the graph below.



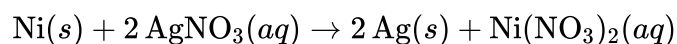
(h) A 1000. g sample of a saturated solution at  $100.^{\circ}\text{C}$  is cooled to  $60.^{\circ}\text{C}$ . Determine the mass of  $\text{AgCl}(s)$  that would precipitate out of the solution.

(i) The specific heat capacity of the solution is  $3.4\text{ J}/(\text{g} \cdot ^{\circ}\text{C})$ . How much heat is released when 1000. g of the solution is cooled from  $100.^{\circ}\text{C}$  to  $60.^{\circ}\text{C}$ ?

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(j) The student finds that AgCl is more soluble in NaCl(aq) than in water because of the series of reactions represented above. Determine the value of  $K_3$  for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic nickel, Ni(s), in a solution of AgNO<sub>3</sub>(aq) and observes that the following reaction occurs.



(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the Ni(s) decreases significantly as it sits in the AgNO<sub>3</sub> solution. The student claims that the mass decreases because the atoms in Ni(s) lose electrons as the experiment progresses. Does the student's claim completely explain the decrease in the mass of the Ni(s)? Justify your answer.

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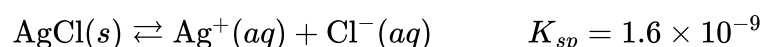
10. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of  $\text{Ag}^+(aq)$  in a solution of  $\text{AgNO}_3(aq)$ . The student adds an excess of  $4.0\text{ M HCl}(aq)$  to  $400.0\text{ mL}$  of the solution, causing  $\text{AgCl}(s)$  to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

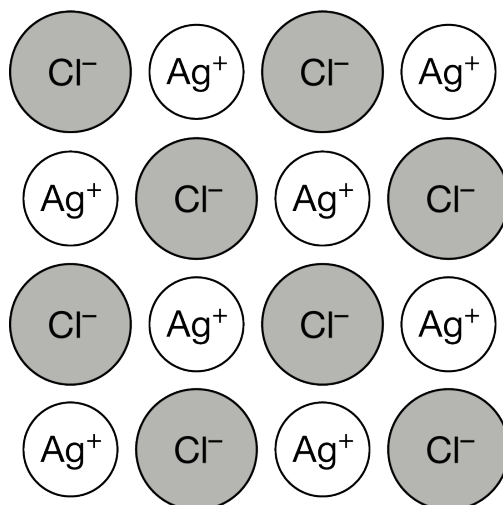
|                                                                  |         |
|------------------------------------------------------------------|---------|
| Initial mass of filter paper                                     | 0.821 g |
| Mass of filter paper + $\text{AgCl}(s)$ precipitate after drying | 1.110 g |

- (a) Calculate the number of moles of  $\text{AgCl}(s)$  that precipitated. (The molar mass of  $\text{AgCl}$  is  $143\text{ g/mol}$ ).
- (b) Assuming that all of the  $\text{Ag}^+(aq)$  ions precipitated, calculate the original molarity of  $\text{Ag}^+(aq)$  in the solution before  $\text{HCl}(aq)$  was added.
- (c) The calculated concentration of  $\text{Ag}^+(aq)$  from the student's data is higher than the actual concentration of the original solution. The student claims that the calculated concentration is too high because the filter paper and  $\text{AgCl}(s)$  precipitate were not fully dry. Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate,  $\text{AgCl}(s)$ , is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of  $\text{AgCl}(s)$  in water and a  $K_{sp}$  value at the temperature of the water.

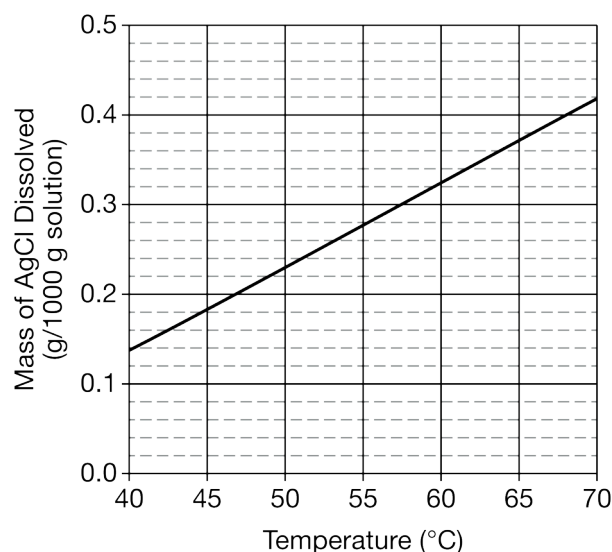


- (d) Calculate  $[\text{Cl}^-]$  in the student's solution.
- (e) The value of  $K_{sp}$  of another chloride salt,  $\text{CuCl}(s)$ , is  $1.7 \times 10^{-7}$ . If separate, saturated solutions of  $\text{CuCl}$  and  $\text{AgCl}$  are prepared, which solution will have the greater  $[\text{Cl}^-]$ ? Justify your answer.
- (f) The student finds the following particulate model of  $\text{AgCl}(s)$ . Assuming the crystal structures are similar, how should the student modify the model to represent  $\text{CuCl}(s)$ ? Justify your answer in terms of the radius and arrangement of electrons in the ions.

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(g) The enthalpy of dissolution of  $\text{AgCl}(s)$  in water is a positive value. Based on this information, the student claims that the energy required to separate the ions in  $\text{AgCl}(s)$  is greater than the energy absorbed during the hydration of  $\text{Ag}^+$  and  $\text{Cl}^-$  by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

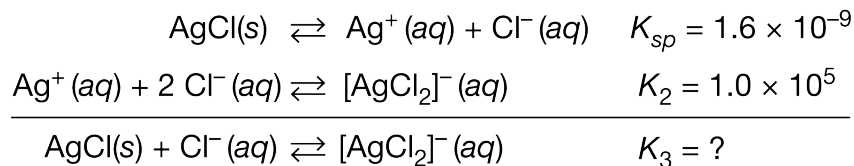
The student then conducts an experiment to determine the solubility of  $\text{AgCl}$  in a  $2.5\text{ M NaCl}$  solution at various temperatures. Some  $\text{AgCl}(s)$  remains in the container at all temperatures studied. The data are plotted on the graph below.



(h) A  $1000.\text{ g}$  sample of a saturated solution at  $50.^{\circ}\text{C}$  is heated to  $70.^{\circ}\text{C}$ . Determine the additional mass of  $\text{AgCl}(s)$  that would dissolve into the solution.

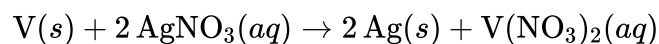
(i) The specific heat capacity of the solution is  $3.3\text{ J}/(\text{g} \cdot ^{\circ}\text{C})$ . How much heat is absorbed when  $1000.\text{ g}$  of the solution is heated from  $50.^{\circ}\text{C}$  to  $70.^{\circ}\text{C}$ ?



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(j) The student finds that AgCl is more soluble in NaCl(aq) than in water because of the series of reactions represented above. Determine the value of  $K_3$  for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic vanadium, V(s), in a solution of AgNO<sub>3</sub>(aq) and observes that the following reaction occurs.



(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the V(s) decreases significantly as it sits in the AgNO<sub>3</sub> solution. The student claims that the mass decreases because the atoms in the V(s) lose protons as the experiment progresses. Does the student's claim accurately explain the decrease in the mass of the V(s)? Justify your answer.

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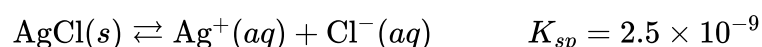
11. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates various properties of silver compounds in the lab. First, the student carries out the following procedure to determine the concentration of  $\text{Ag}^+(aq)$  in a solution of  $\text{AgNO}_3(aq)$ . The student adds an excess of  $6.0\text{ M HCl}(aq)$  to  $500.0\text{ mL}$  of the solution, causing  $\text{AgCl}(s)$  to precipitate. The precipitate is separated from the solution by filtering it through a pre-weighed piece of filter paper. The precipitate and filter paper are rinsed with distilled water, dried, and weighed. The data from the experiment are given in the table below.

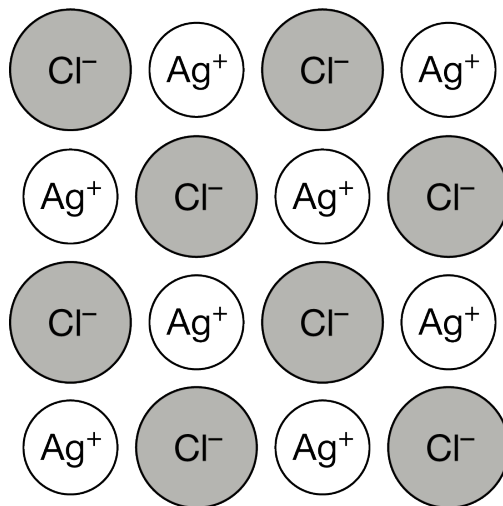
|                                                                  |         |
|------------------------------------------------------------------|---------|
| Initial mass of filter paper                                     | 0.945 g |
| Mass of filter paper + $\text{AgCl}(s)$ precipitate after drying | 3.095 g |

- (a) Calculate the number of moles of  $\text{AgCl}(s)$  that precipitated. (The molar mass of  $\text{AgCl}$  is  $143\text{ g/mol}$ ).
- (b) Assuming that all of the  $\text{Ag}^+(aq)$  ions precipitated, calculate the original molarity of  $\text{Ag}^+(aq)$  in the solution before  $\text{HCl}(aq)$  was added.
- (c) The calculated concentration of  $\text{Ag}^+(aq)$  from the student's data is higher than the actual concentration of the original solution. The student claims that the calculated concentration is too high because  $8.0\text{ M HCl}(aq)$  was accidentally used instead of  $6.0\text{ M HCl}(aq)$ . Do you agree or disagree? Justify your answer.

After the experiment, the student learns that the precipitate,  $\text{AgCl}(s)$ , is slightly soluble in water. To determine the solubility, the student mixes a sample of the precipitate with warm, distilled water to make a saturated solution. The student finds the following equation for the dissolution of  $\text{AgCl}(s)$  in water and a  $K_{sp}$  value at the temperature of the water.

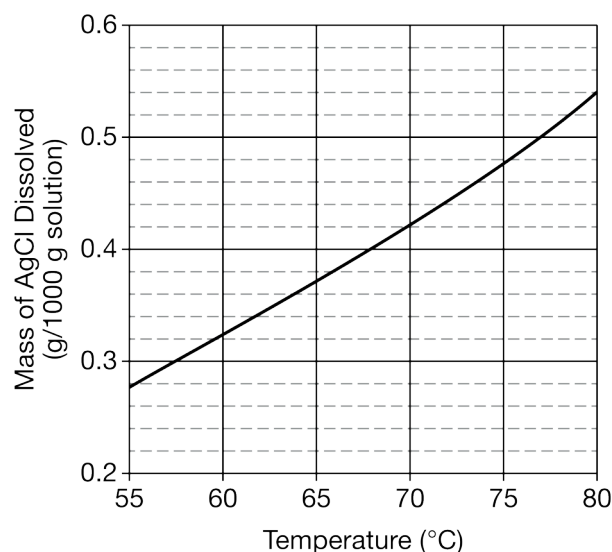


- (d) Calculate  $[\text{Cl}^-]$  in the student's solution.
- (e) The value of  $K_{sp}$  of another chloride salt,  $\text{LiCl}(s)$ , is  $4.0 \times 10^2$ . If separate, saturated solutions of  $\text{LiCl}$  and  $\text{AgCl}$  are prepared, which solution will have the greater  $[\text{Cl}^-]$ ? Justify your answer.
- (f) The student finds the following particulate model of  $\text{AgCl}(s)$ . Assuming the crystal structures are similar, how should the student modify the model to represent  $\text{LiCl}(s)$ ? Justify your answer in terms of the radius and arrangement of electrons in the ions.

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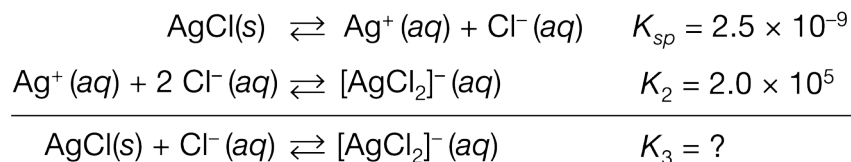
(g) The enthalpy of dissolution of  $\text{AgCl}(s)$  in water is a positive value. Based on this information, the student claims that the energy released during the separation of the ions in  $\text{AgCl}(s)$  is greater than the energy absorbed during the hydration of  $\text{Ag}^+$  and  $\text{Cl}^-$  by water molecules. Do you agree or disagree with the student's claim? Justify your answer.

The student then conducts an experiment to determine the solubility of  $\text{AgCl}$  in a  $2.5\text{ M NaCl}$  solution at various temperatures. Some  $\text{AgCl}(s)$  remains in the container at all temperatures studied. The data are plotted on the graph below.



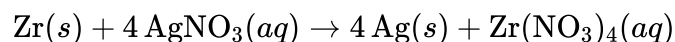
(h) A 1000. g sample of a saturated solution at  $60.^{\circ}\text{C}$  is heated to  $75.^{\circ}\text{C}$ . Determine the additional mass of  $\text{AgCl}(s)$  that would dissolve into the solution.

(i) The specific heat capacity of the solution is  $3.4\text{ J}/(\text{g} \cdot ^{\circ}\text{C})$ . How much heat is absorbed when 1000. g of the solution is heated from  $60.^{\circ}\text{C}$  to  $75.^{\circ}\text{C}$ ?

Chapter 10 Part 3: K<sub>sp</sub>

(j) The student finds that AgCl is more soluble in NaCl(aq) than in water because of the series of reactions above. Determine the value of  $K_3$  for the overall reaction.

To further investigate the reactivity of silver, the student submerges a piece of metallic zirconium, Zr(s), in a solution of AgNO<sub>3</sub>(aq) and observes that the following reaction occurs.



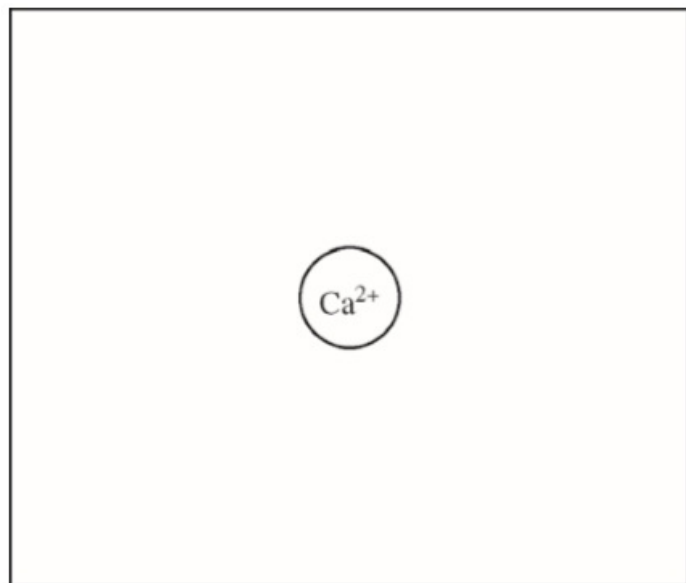
(k) Write the balanced net-ionic equation for the reaction.

(l) The mass of the Zr(s) decreases significantly as it sits in the AgNO<sub>3</sub> solution. The student claims that the mass decreases because the Zr(s) melts into a liquid as the experiment progresses. Does the student's claim accurately explain the decrease in the mass of the Zr(s)? Justify your answer.

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12. Answer the following questions about the solubility of  $\text{Ca(OH)}_2$  ( $K_{sp} = 1.3 \times 10^{-6}$ ).
- Write a balanced chemical equation for the dissolution of  $\text{Ca(OH)}_2(s)$  in pure water.
  - Calculate the molar solubility of  $\text{Ca(OH)}_2$  in  $0.10\text{ M Ca(NO}_3)_2$ .
  - In the box below, complete a particle representation diagram that includes four water molecules with proper orientation around the  $\text{Ca}^{2+}$  ion.

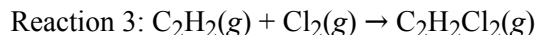
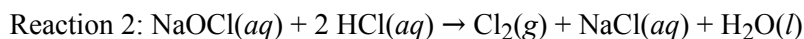
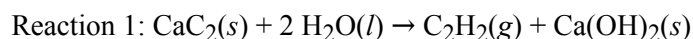
Represent water molecules as .



13. Answer the following questions about the solubility of  $\text{AgCl}(s)$ . The value of  $K_{sp}$  for  $\text{AgCl}(s)$  is  $1.8 \times 10^{-10}$ .
- Calculate the value of  $[\text{Ag}^+]$  in a saturated solution of  $\text{AgCl}$  in distilled water.
  - The concentration of  $\text{Cl}^-(aq)$  in seawater is  $0.54\text{ M}$ .
    - Calculate the molar solubility of  $\text{AgCl}(s)$  in seawater.
    - Explain why  $\text{AgCl}(s)$  is less soluble in seawater than in distilled water.

**Chapter 10 Part 3: K<sub>sp</sub>**

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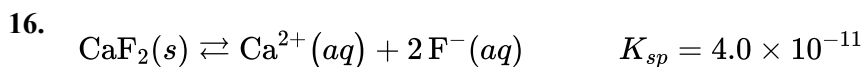
14.  $\text{Ca(OH)}_2(s)$  precipitates when a 1.0 g sample of  $\text{CaC}_2(s)$  is added to 1.0 L of distilled water at room temperature. If a 0.064 g sample of  $\text{CaC}_2(s)$  (molar mass 64 g/mol) is used instead and all of it reacts, which of the following will occur and why? (The value of  $K_{sp}$  for  $\text{Ca(OH)}_2$  is  $8.0 \times 10^{-8}$ .)
- (A)  $\text{Ca(OH)}_2$  will precipitate because  $Q > K_{sp}$  .
- (B)  $\text{Ca(OH)}_2$  will precipitate because  $Q < K_{sp}$  .
- (C)  $\text{Ca(OH)}_2$  will not precipitate because  $Q > K_{sp}$  .
- (D)  $\text{Ca(OH)}_2$  will not precipitate because  $Q < K_{sp}$  .
- 



$$\Delta H > 0$$

Dissolution of the slightly soluble salt  $\text{CaF}_2$  is shown by the equation above. Which of the following changes will decrease  $[\text{Ca}^{2+}]$  in a saturated solution of  $\text{CaF}_2$  , and why? (Assume that after each change some  $\text{CaF}_2(s)$  remains in contact with the solution.)

- (A) Allowing some of the water to evaporate from the solution, because more  $\text{CaF}_2(s)$  will precipitate
- (B) Adding 0.1 M  $\text{HNO}_3(aq)$  , because some  $\text{F}^-(aq)$  ions will become protonated
- (C) Adding 0.1 M  $\text{NaNO}_3(aq)$  , because additional liquid will dilute the solution
- (D) Adding  $\text{NaF}(s)$ , because the reaction will proceed toward reactants

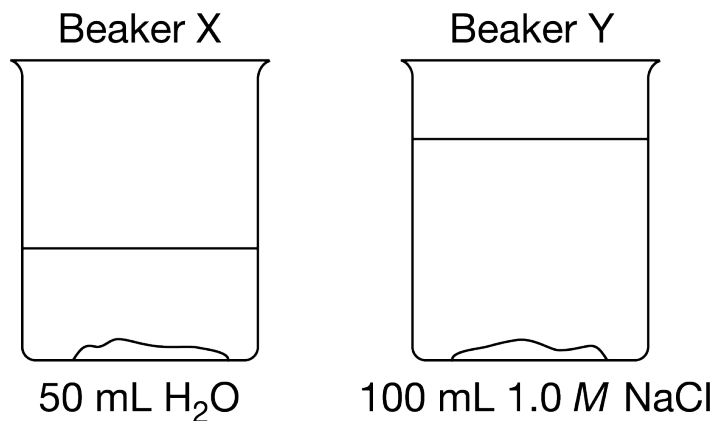


The concentration of  $\text{F}^-(aq)$  in drinking water that is considered to be ideal for promoting dental health is  $4.0 \times 10^{-5} \text{ M}$ . Based on the information above, the maximum concentration of  $\text{Ca}^{2+}(aq)$  that can be present in drinking water without lowering the concentration of  $\text{F}^-(aq)$  below the ideal level is closest to

- (A) 0.25 M
- (B) 0.025 M
- (C)  $1.6 \times 10^{-6} \text{ M}$
- (D)  $1.6 \times 10^{-15} \text{ M}$
-

Chapter 10 Part 3: K<sub>sp</sub>

17.



Beaker X contains 50 mL of distilled water and beaker Y contains 100 mL of 1.0 *M* NaCl. Solid AgCl is added to each of the beakers. After thoroughly stirring the contents of the beakers, some solid AgCl remains at the bottom of each beaker, as shown above. Which of the following is true?

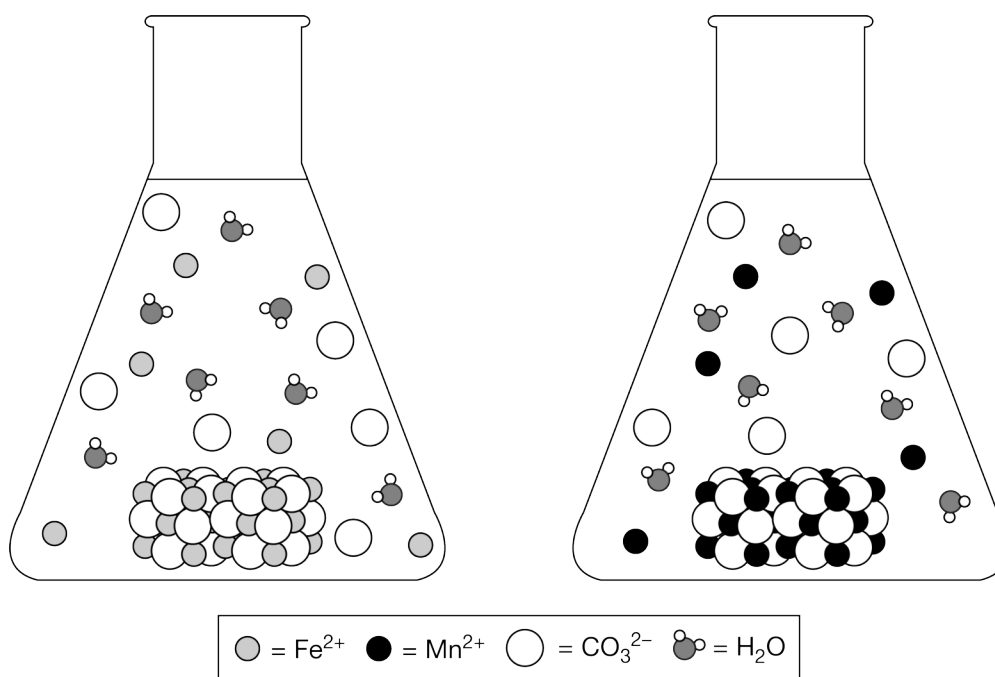
- (A)  $[\text{Ag}^+]$  is zero in both beakers.
- (B)  $[\text{Ag}^+]$  is the same, but not zero, in both beakers.
- (C)  $[\text{Ag}^+]$  is greater in beaker X.
- (D)  $[\text{Ag}^+]$  is greater in beaker Y.

Chapter 10 Part 3: K<sub>sp</sub>

18.

| Reaction                                                                        | $K_{sp}$            | $\Delta H^\circ$ | $\Delta S^\circ$ |
|---------------------------------------------------------------------------------|---------------------|------------------|------------------|
| $\text{FeCO}_3(s) \rightleftharpoons \text{Fe}^{2+}(aq) + \text{CO}_3^{2-}(aq)$ | $3 \times 10^{-11}$ | $< 0$            | $> 0$            |
| $\text{MnCO}_3(s) \rightleftharpoons \text{Mn}^{2+}(aq) + \text{CO}_3^{2-}(aq)$ | $2 \times 10^{-11}$ | $< 0$            | $> 0$            |

The table above lists the equilibrium constants and changes in thermodynamic properties for the dissolution of  $\text{FeCO}_3$  and  $\text{MnCO}_3$  at  $25^\circ\text{C}$ . The two particle diagrams below represent saturated solutions of each compound at equilibrium.



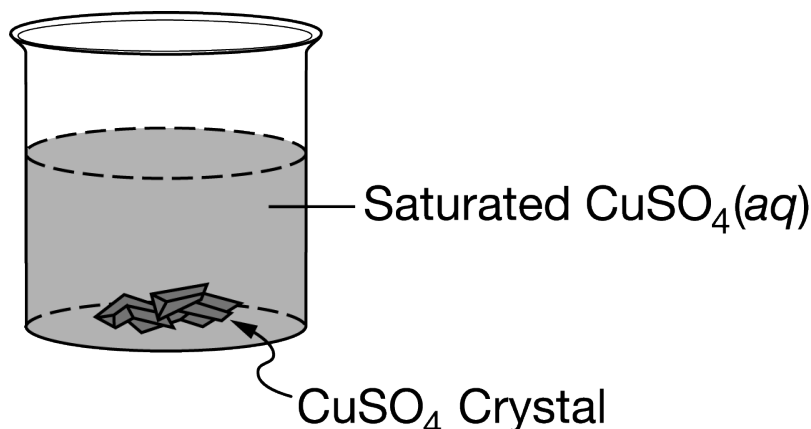
Which of the following explains which of the properties listed in the table is best represented by the particle diagram?

- (A) The particle diagrams best represent that  $\Delta H^\circ < 0$  because the ions from both compounds are solvated by water molecules.
- (B) The particle diagrams best represent that  $\Delta H^\circ < 0$  because both compounds produce about the same amount of  $\text{CO}_3^{2-}$  ions from the dissolution.
- (C) The particle diagrams best represent that  $\Delta S^\circ > 0$  because both compounds produce a very small amount of ions from the dissolution.
- (D) The particle diagrams best represent that the molar solubility is greater for  $\text{FeCO}_3$  compared to  $\text{MnCO}_3$ .



Chapter 10 Part 3: K<sub>sp</sub>

19.



The saturated  $\text{CuSO}_4(aq)$  shown above is left uncovered on a lab bench at a constant temperature. As the solution evaporates, 1.0 mL samples of the solution are removed every three days and the  $[\text{SO}_4^{2-}]$  in the samples is measured. It is observed that the  $[\text{SO}_4^{2-}]$  in the solution did not change over time. Which of the following best helps to explain the observation?

- (A) As the solution evaporates,  $\text{Cu}^{2+}(aq)$  and  $\text{SO}_4^{2-}(aq)$  leave the beaker along with the water molecules.
- (B) As the solution evaporates, the dissolution of  $\text{CuSO}_4(s)$  in the beaker decreases.
- (C) The evaporation of water is endothermic, so more  $\text{CuSO}_4(s)$  dissolves exothermically in the solution, which increases the  $[\text{SO}_4^{2-}]$ .
- (D) As water evaporates, more  $\text{CuSO}_4(s)$  precipitates out of the solution in the beaker.

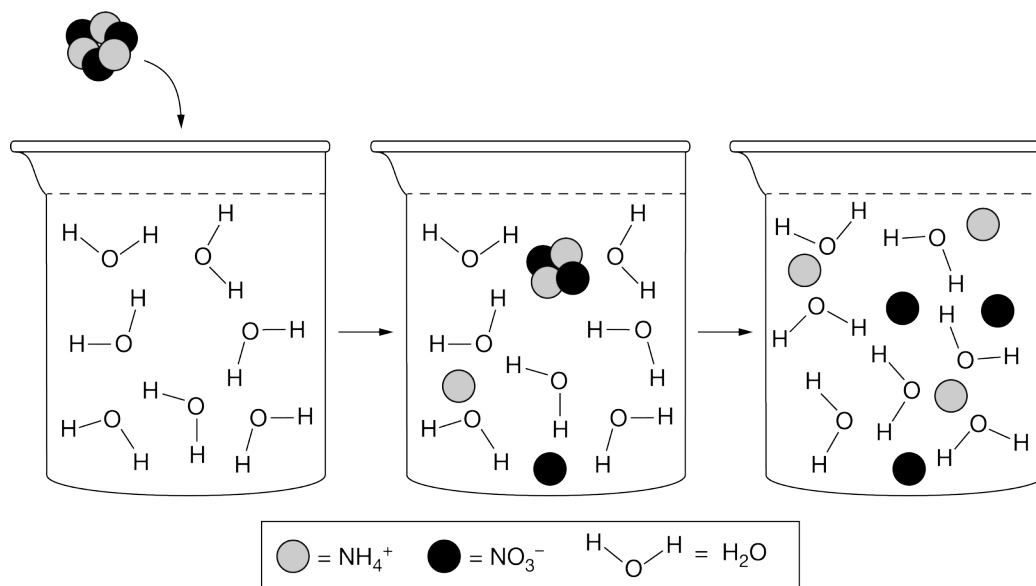
20.  $\text{CdF}_2(s) \rightleftharpoons \text{Cd}^{2+}(aq) + 2 \text{F}^{-}(aq)$ 

A saturated aqueous solution of  $\text{CdF}_2$  is prepared. The equilibrium in the solution is represented above. In the solution,  $[\text{Cd}^{2+}]_{eq} = 0.0585 M$  and  $[\text{F}^{-}]_{eq} = 0.117 M$ . Some  $0.90 M$   $\text{NaF}$  is added to the saturated solution. Which of the following identifies the molar solubility of  $\text{CdF}_2$  in pure water and explains the effect that the addition of  $\text{NaF}$  has on this solubility?

- (A) The molar solubility of  $\text{CdF}_2$  in pure water is  $0.0585 M$ , and adding  $\text{NaF}$  decreases this solubility because the equilibrium shifts to favor the precipitation of some  $\text{CdF}_2$ .
- (B) The molar solubility of  $\text{CdF}_2$  in pure water is  $0.0585 M$ , and adding  $\text{NaF}$  has no effect on the solubility because only changes in temperature can increase or decrease the molar solubility of an ionic solid.
- (C) The molar solubility of  $\text{CdF}_2$  in pure water is  $0.117 M$ , and adding  $\text{NaF}$  decreases this solubility because the equilibrium shifts to favor the precipitation of some  $\text{CdF}_2$ .
- (D) The molar solubility of  $\text{CdF}_2$  in pure water is  $0.176 M$ , and adding  $\text{NaF}$  increases this solubility because the  $\text{Na}^{+}$  ions displace the  $\text{Cd}^{2+}$  ions, causing the equilibrium to shift to favor the products.

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21.



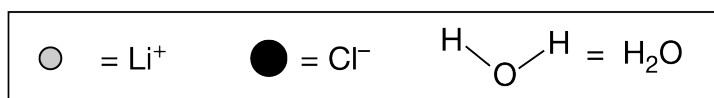
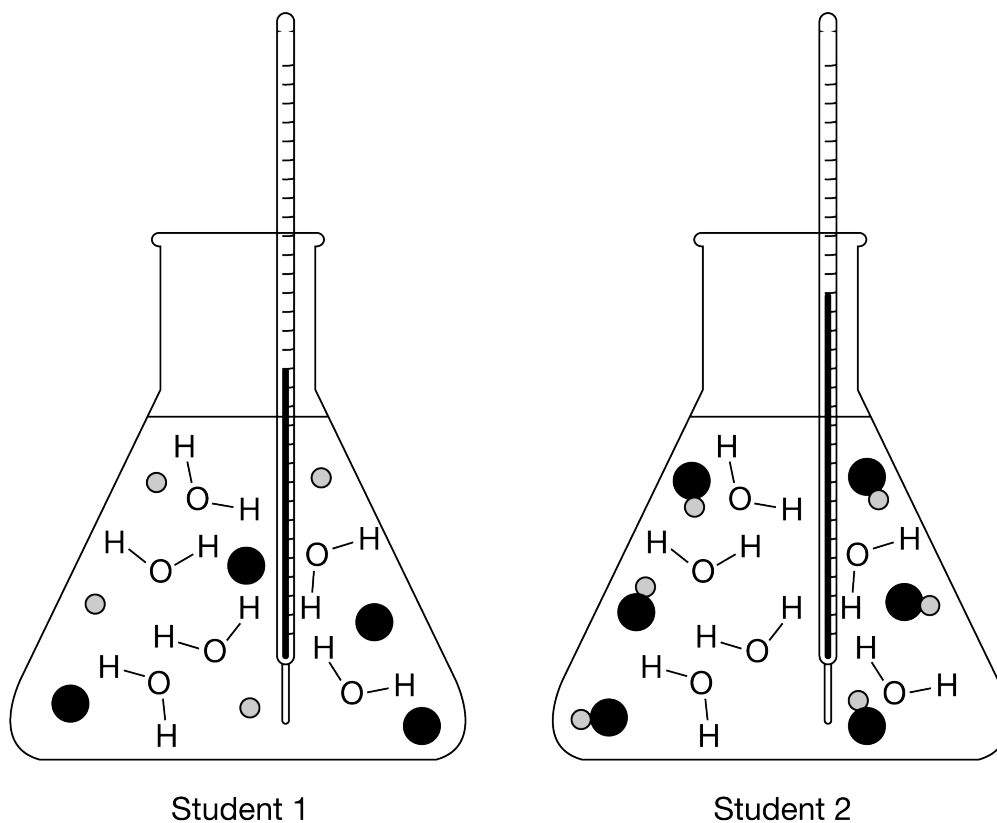
At 298 K,  $\text{NH}_4\text{NO}_3$  readily dissolves in water, suggesting that the change in free energy ( $\Delta G$ ) favors the dissolution process. However, when  $\text{NH}_4\text{NO}_3$  dissolves in water, the temperature of the water decreases. The particulate diagram above attempts to provide a microscopic view of the dissolution of  $\text{NH}_4\text{NO}_3(s)$  considering both the change in enthalpy ( $\Delta H$ ) and the change in entropy ( $\Delta S$ ). Which of the following explains what the particle diagram is able to illustrate and why?

- (A) The particle diagram is able to illustrate that the  $\Delta H$  for the dissolution of  $\text{NH}_4\text{NO}_3$  is exothermic because overcoming the attractive forces between the ions requires the absorption of energy.
- (B) The particle diagram is able to illustrate that the  $\Delta H$  for the dissolution of  $\text{NH}_4\text{NO}_3$  is endothermic because forming new interactions between the ions and the water molecules releases energy.
- (C) The particle diagram is able to illustrate that entropy increases when  $\text{NH}_4\text{NO}_3(s)$  dissolves in water because the ions disperse in solution.
- (D) The particle diagram is able to illustrate that entropy decreases when  $\text{NH}_4\text{NO}_3(s)$  dissolves in water because many hydrogen bonds can form between the water molecules and the ions.

Chapter 10 Part 3: K<sub>sp</sub>

22.

| Property                                                  | Student 1 | Student 2 |
|-----------------------------------------------------------|-----------|-----------|
| Mass of LiCl( <i>g</i> )                                  | 2.0       | 3.0       |
| Mass of H <sub>2</sub> O( <i>g</i> )                      | 100.0     | 100.0     |
| Initial temperature of H <sub>2</sub> O(°C)               | 20.0      | 20.1      |
| Temperature of H <sub>2</sub> O after LiCl dissolved (°C) | 23.9      | 26.1      |



Two students prepared aqueous solutions of LiCl and measured the properties, as shown in the table above. Both students observed that the solid LiCl readily dissolved in H<sub>2</sub>O. The students drew particle diagrams to explain the changes in the enthalpy and entropy of dissolution for LiCl based on their results and observations. Based on this information, the better particle diagram was drawn by which student, and why is that diagram more accurate?

Chapter 10 Part 3:  $K_{sp}$ 

- (A) The better particle diagram was drawn by Student 1 because when  $\text{LiCl}$  dissolves in water, it dissociates into  $\text{Li}^+$  and  $\text{Cl}^-$  ions causing an increase in entropy.
- (B) The better particle diagram was drawn by Student 1 because when  $\text{LiCl}$  dissolves in water, it dissociates into  $\text{Li}^+$  and  $\text{Cl}^-$  ions indicating that the dissolution of  $\text{LiCl}$  is exothermic.
- (C) The better particle diagram was drawn by Student 2 because it shows that  $\text{LiCl}$  does not dissociate, resulting in a decrease in entropy for the dissolution of  $\text{LiCl}$ .
- (D) The better particle diagram was drawn by Student 2 because it shows that  $\text{LiCl}$  does not dissociate, resulting in an endothermic process for the dissolution of  $\text{LiCl}$ .

23.

| Compound                    | $K_{sp}$ at $25^\circ\text{C}$ |
|-----------------------------|--------------------------------|
| $\text{Pb}(\text{OH})_2(s)$ | $1.2 \times 10^{-15}$          |
| $\text{PbI}_2(s)$           | $1.4 \times 10^{-8}$           |
| $\text{PbF}_2(s)$           | $4.0 \times 10^{-8}$           |

Equimolar samples of  $\text{Pb}(\text{OH})_2(s)$ ,  $\text{PbI}_2(s)$ , and  $\text{PbF}_2(s)$  are placed in three separate beakers, each containing 250 mL of water at  $25^\circ\text{C}$ . After the solutions are stirred, solid remains in the bottom of each beaker. Based on the  $K_{sp}$  values for the compounds listed in the table above, a solution of which of the compounds will have the lowest  $[\text{Pb}^{2+}]$ ?

- (A)  $\text{Pb}(\text{OH})_2(s)$
- (B)  $\text{PbI}_2(s)$  only
- (C)  $\text{PbF}_2(s)$  only
- (D) Both  $\text{PbI}_2(s)$  and  $\text{PbF}_2(s)$  will produce solutions with the same, lowest  $[\text{Pb}^{2+}]$ .

24.

| Compound                 | $K_{sp}$              |
|--------------------------|-----------------------|
| $\text{PbCl}_2$          | $1.2 \times 10^{-5}$  |
| $\text{CuCl}$            | $1.6 \times 10^{-7}$  |
| $\text{AgCl}$            | $1.8 \times 10^{-10}$ |
| $\text{Hg}_2\text{Cl}_2$ | $1.4 \times 10^{-18}$ |

Based on the  $K_{sp}$  values in the table above, a saturated solution of which of the following compounds has the highest  $[\text{Cl}^-]$ ?

**Chapter 10 Part 3:  $K_{sp}$** 

- (A)  $\text{PbCl}_2$   
(B)  $\text{CuCl}$   
(C)  $\text{AgCl}$   
(D)  $\text{Hg}_2\text{Cl}_2$
25. High solubility of an ionic solid in water is favored by which of the following conditions?
- I. The existence of strong ionic attractions in the crystal lattice  
II. The formation of strong ion-dipole attractions  
III. An increase in entropy upon dissolving
- (A) I only  
(B) I and II only  
(C) I and III only  
(D) II and III only  
(E) I, II, and III
26. How many moles of NaF must be dissolved in 1.00 liter of a saturated solution of  $\text{PbF}_2$  at  $25^\circ\text{C}$  to reduce the  $[\text{Pb}^{2+}]$  to  $1 \times 10^{-6}$  molar?
- ( $K_{sp} \text{ PbF}_2 \text{ at } 25^\circ\text{C} = 4.0 \times 10^{-8}$ )
- (A) 0.020 mole  
(B) 0.040 mole  
(C) 0.10 mole  
(D) 0.20 mole  
(E) 0.40 mole
27. In a saturated solution of  $\text{Zn}(\text{OH})_2$  at  $25^\circ\text{C}$ , the value of  $[\text{OH}^-]$  is  $2.0 \times 10^{-6} \text{ M}$ . What is the value of the solubility-product constant,  $K_{sp}$ , for  $\text{Zn}(\text{OH})_2$  at  $25^\circ\text{C}$ ?
- (A)  $4.0 \times 10^{-18}$   
(B)  $8.0 \times 10^{-18}$   
(C)  $1.6 \times 10^{-17}$   
(D)  $4.0 \times 10^{-12}$   
(E)  $2.0 \times 10^{-6}$
28. The value of  $K_{sp}$  for the salt  $\text{PbF}_2$  is  $3.2 \times 10^{-8}$ . The  $[\text{Pb}^{2+}]$  in a saturated solution of  $\text{PbF}_2$  is approximately
- (A)  $2 \times 10^{-4} \text{ M}$   
(B)  $4 \times 10^{-4} \text{ M}$   
(C)  $2 \times 10^{-3} \text{ M}$   
(D)  $4 \times 10^{-3} \text{ M}$

Chapter 10 Part 3: K<sub>sp</sub>

29.

|                                                        |         |
|--------------------------------------------------------|---------|
| Mass of KI tablet                                      | 0.425 g |
| Mass of thoroughly dried filter paper                  | 1.462 g |
| Mass of filter paper + precipitate after first drying  | 1.775 g |
| Mass of filter paper + precipitate after second drying | 1.699 g |
| Mass of filter paper + precipitate after third drying  | 1.698 g |

A student is given the task of determining the I<sup>−</sup> content of tablets that contain KI and an inert, water-soluble sugar as a filler. A tablet is dissolved in 50.0 mL of distilled water, and an excess of 0.20 M Pb(NO<sub>3</sub>)<sub>2</sub>(aq) is added to the solution. A yellow precipitate forms, which is then filtered, washed, and dried. The data from the experiment are shown in the table above.

- For the chemical reaction that occurs when the precipitate forms,
  - write a balanced, net-ionic equation for the reaction, and
  - explain why the reaction is best represented by a net-ionic equation.
- Explain the purpose of drying and weighing the filter paper with the precipitate three times.
- In the filtrate solution, is [K<sup>+</sup>] greater than, less than, or equal to [NO<sub>3</sub><sup>−</sup>] ? Justify your answer.
- Calculate the number of moles of precipitate that is produced in the experiment.
- Calculate the mass percent of I<sup>−</sup> in the tablet.
- In another trial, the student dissolves a tablet in 55.0 mL of water instead of 50.0 mL of water. Predict whether the experimentally determined mass percent of I<sup>−</sup> will be greater than, less than, or equal to the amount calculated in part (e). Justify your answer.
- A student in another lab also wants to determine the I<sup>−</sup> content of a KI tablet but does not have access to Pb(NO<sub>3</sub>)<sub>2</sub>. However, the student does have access to 0.20 M AgNO<sub>3</sub>, which reacts with I<sup>−</sup>(aq) to produce AgI(s). The value of K<sub>sp</sub> for AgI is 8.5 × 10<sup>−17</sup>.
  - Will the substitution of AgNO<sub>3</sub> for Pb(NO<sub>3</sub>)<sub>2</sub> result in the precipitation of the I<sup>−</sup> ion from solution? Justify your answer.
  - The student only has access to one KI tablet and a balance that can measure to the nearest 0.01 g. Will the student be able to determine the mass of AgI produced to three significant figures? Justify your answer.

Chapter 10 Part 3: K<sub>sp</sub>

30.

|                                                        |         |
|--------------------------------------------------------|---------|
| Mass of KI tablet                                      | 0.425 g |
| Mass of thoroughly dried filter paper                  | 1.462 g |
| Mass of filter paper + precipitate after first drying  | 1.775 g |
| Mass of filter paper + precipitate after second drying | 1.699 g |
| Mass of filter paper + precipitate after third drying  | 1.698 g |

A student is given the task of determining the  $\text{I}^-$  content of tablets that contain KI and an inert, water-soluble sugar as a filler. A tablet is dissolved in 50.0 mL of distilled water, and an excess of  $0.20\text{ M Pb(NO}_3)_2(\text{aq})$  is added to the solution. A yellow precipitate forms, which is then filtered, washed, and dried. The data from the experiment are shown in the table above.

- For the chemical reaction that occurs when the precipitate forms,
  - write a balanced, net-ionic equation for the reaction, and
  - explain why the reaction is best represented by a net-ionic equation.
- Explain the purpose of drying and weighing the filter paper with the precipitate three times.
- In the filtrate solution, is  $[\text{K}^+]$  greater than, less than, or equal to  $[\text{NO}_3^-]$ ? Justify your answer.
- Calculate the number of moles of precipitate that is produced in the experiment.
- Calculate the mass percent of  $\text{I}^-$  in the tablet.
- In another trial, the student dissolves a tablet in 55.0 mL of water instead of 50.0 mL of water. Predict whether the experimentally determined mass percent of  $\text{I}^-$  will be greater than, less than, or equal to the amount calculated in part (e). Justify your answer.
- A student in another lab also wants to determine the  $\text{I}^-$  content of a KI tablet but does not have access to  $\text{Pb(NO}_3)_2$ . However, the student does have access to  $0.20\text{ M AgNO}_3$ , which reacts with  $\text{I}^-(\text{aq})$  to produce  $\text{AgI(s)}$ . The value of  $K_{\text{sp}}$  for  $\text{AgI}$  is  $8.5 \times 10^{-17}$ .
  - Will the substitution of  $\text{AgNO}_3$  for  $\text{Pb(NO}_3)_2$  result in the precipitation of the  $\text{I}^-$  ion from solution? Justify your answer.
  - The student only has access to one KI tablet and a balance that can measure to the nearest 0.01 g. Will the student be able to determine the mass of  $\text{AgI}$  produced to three significant figures?

Chapter 10 Part 3:  $K_{sp}$ 

Justify your answer.

31. The value of  $K_{sp}$  for  $PbCl_2$  is  $1.6 \times 10^{-5}$ . What is the lowest concentration of  $Cl^-(aq)$  that would be needed to begin precipitation of  $PbCl_2(s)$  in  $0.010\ M\ Pb(NO_3)_2$ ?

- (A)  $1.6 \times 10^{-7}\ M$   
 (B)  $4.0 \times 10^{-4}\ M$   
 (C)  $1.6 \times 10^{-3}\ M$   
 (D)  $2.6 \times 10^{-3}\ M$   
 (E)  $4.0 \times 10^{-2}\ M$

32.

| Compound           | $[Pb^{2+}]$ in Saturated Solution |
|--------------------|-----------------------------------|
| PbS                | $1.7 \times 10^{-14}\ M$          |
| PbCO <sub>3</sub>  | $2.7 \times 10^{-7}\ M$           |
| PbCrO <sub>4</sub> | $5.5 \times 10^{-7}\ M$           |
| PbSO <sub>4</sub>  | $1.6 \times 10^{-4}\ M$           |

Based on  $[Pb^{2+}]$  in saturated solutions of the compounds listed in the table above, which of the compounds has the smallest  $K_{sp}$  value?

- (A) PbS  
 (B) PbCO<sub>3</sub>  
 (C) PbCrO<sub>4</sub>  
 (D) PbSO<sub>4</sub>

33.  $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq) \quad K_{sp} = 1.8 \times 10^{-10}$

Shown above is information about the dissolution of  $AgCl(s)$  in water at 298 K. In a chemistry lab a student wants to determine the value of  $s$ , the molar solubility of  $AgCl$ , by measuring  $[Ag^+]$  in a saturated solution prepared by mixing excess  $AgCl$  and distilled water. How would the results of the experiment be altered if the student mixed excess  $AgCl$  with tap water (in which  $[Cl^-] = 0.010\ M$ ) instead of distilled water and the student did not account for the  $Cl^-$  in the tap water?

- (A) The value obtained for  $K_{sp}$  would be too small because  $Cl^-(aq)$  ions would be attracted to the  $Ag^+$  ions in the  $AgCl$  crystals, thus preventing water molecules from reaching the crystals.  
 (B) The value obtained for  $K_{sp}$  would be too small because less  $AgCl(s)$  would dissolve because of the common ion effect due to the  $Cl^-(aq)$  already in the water.  
 (C) The value obtained for  $K_{sp}$  would be too large because more  $AgCl(s)$  would dissolve because of the attractions between  $Ag^+$  ions in the  $AgCl$  crystals and the  $Cl^-(aq)$  ions in the water.  
 (D) The results of the experiment would not be altered because  $0.010\ M$  is such a small concentration of  $Cl^-(aq)$  ions and thus has no effect on the dissolution of  $AgCl(s)$ .



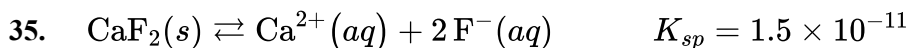
Chapter 10 Part 3:  $K_{sp}$ 

34.

| Substance                | $K_{sp}$                               |
|--------------------------|----------------------------------------|
| $\text{Sn}(\text{OH})_2$ | $K_{sp} = 4S^3 = 5.45 \times 10^{-27}$ |
| $\text{CuCN}$            | $K_{sp} = S^2 = 3.47 \times 10^{-20}$  |
| $\text{MgF}_2$           | $K_{sp} = 4S^3 = 5.16 \times 10^{-11}$ |
| $\text{NiCO}_3$          | $K_{sp} = S^2 = 1.42 \times 10^{-7}$   |

The equilibrium constants for the dissolution ( $K_{sp}$ ) of various substances in aqueous solution at  $25^\circ\text{C}$  are listed in the table above. Which of the following provides a correct comparison of the molar solubilities ( $S$ ) of some of these substances based on their  $K_{sp}$ ?

- (A) The molar solubilities for  $\text{CuCN}$  and  $\text{NiCO}_3$  are calculated using  $S = \sqrt{K_{sp}}$  and  $\text{NiCO}_3$  has a lower molar solubility than  $\text{CuCN}$ .
- (B) The molar solubilities for  $\text{CuCN}$  and  $\text{NiCO}_3$  are calculated using  $S = \sqrt{K_{sp}}$  and  $\text{CuCN}$  has a lower molar solubility than  $\text{NiCO}_3$ .
- (C) The molar solubilities for  $\text{Sn}(\text{OH})_2$  and  $\text{MgF}_2$  are calculated using  $S = \sqrt{\frac{K_{sp}}{4}}$  and  $\text{MgF}_2$  has a lower molar solubility than  $\text{NiCO}_3$ .
- (D) The molar solubilities for  $\text{Sn}(\text{OH})_2$  and  $\text{MgF}_2$  are calculated using  $S = \sqrt{\frac{K_{sp}}{4}}$  and  $\text{Sn}(\text{OH})_2$  has a lower molar solubility than  $\text{NiCO}_3$ .



The concentration of  $\text{F}^{-}(aq)$  in drinking water that is considered to be ideal for promoting dental health is  $4.0 \times 10^{-5} M$ . Based on the information above, the maximum concentration of  $\text{Ca}^{2+}(aq)$  that can be present in drinking water without lowering the concentration of  $\text{F}^{-}(aq)$  below the ideal level is closest to

- (A)  $0.0023 M$
- (B)  $0.0094 M$
- (C)  $3.8 \times 10^{-7} M$
- (D)  $6.0 \times 10^{-16} M$

Chapter 10 Part 3:  $K_{sp}$ 

36. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

The majority of elements in the periodic table are metals. At 298 K most metals are solid and share similar physical properties.

(a) A student claims that gold is a good conductor of electricity because it is dense. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with wavelength 130 nm cause the ejection of 6 s electrons from gold, but photons with wavelength 105 nm cause the electrons to be ejected with greater energy. A student claims this is because the photons with wavelength 130 nm are higher in energy than photons with wavelength 105 nm. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, X. The student takes a 100.0 g piece of metal X, heats it to 500.0 °C, then places it on a 1000.0 g block of ice at 0.0 °C. The ice partially melts, and the final temperature of the metal, ice, and melted water is 0.0 °C. The student calculates the experimental value of the specific heat capacity of metal X and records it as 0.13 J/(g · °C).

(c) Calculate the magnitude of the energy change of metal X during the experiment.

(d) Calculate the mass of ice melted by metal X. (The enthalpy of fusion of ice is 330 J/g.)

(e) A student claims that the energy transferred to the ice is used to overcome hydrogen bonding between water molecules in the solid ice. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of  $\text{XCl}_2$ , the chloride salt of metal X, and mixes it with 100.0 mL of distilled water to form a saturated solution. The value of  $K_{sp}$  for  $\text{XCl}_2$  is  $4.0 \times 10^{-12}$  at 298 K.

(f) Write the balanced chemical equation for the change that occurs when solid  $\text{XCl}_2$  dissolves in water.

(g) Calculate the concentration of  $\text{Cl}^- (aq)$  ions in the saturated solution.

(h) Will the solubility of  $\text{XCl}_2$  increase, decrease, or remain the same if it is dissolved in 100.0 mL of 0.10 M NaCl instead of distilled water? Explain.

(i) The student gently heats 100.0 mL of the saturated solution of  $\text{XCl}_2$  to a temperature of 315 K. If the enthalpy change of the dissolution process is positive, will the concentration of  $\text{X}^{2+} (aq)$  ions in the solution be greater than, less than, or the same as when the temperature of the solution was 298 K? Justify your answer.

The first five ionization energies for metal X are shown in the following table.

Chapter 10 Part 3: K<sub>sp</sub>

| Ionization Energies of Metal X (kJ/mol) |        |       |        |       |
|-----------------------------------------|--------|-------|--------|-------|
| First                                   | Second | Third | Fourth | Fifth |
| 590                                     | 1100   | 4900  | 6500   | 8200  |

(j) The ionic compound  $\text{XCl}_3$  is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for  $\text{XF}_2$  and for  $\text{ZF}_2$ . (Z is another metal different from metal X.)

| Dissociation Reaction                                            | Lattice Energy (kJ/mol) |
|------------------------------------------------------------------|-------------------------|
| $\text{XF}_2(s) \rightarrow \text{X}^{2+}(g) + 2\text{F}^{-}(g)$ | 2609                    |
| $\text{ZF}_2(s) \rightarrow \text{Z}^{2+}(g) + 2\text{F}^{-}(g)$ | 2341                    |

(k) Using the information in the table above, predict whether the ionic radius of  $\text{X}^{2+}$  is greater than, less than, or equal to the ionic radius of  $\text{Z}^{2+}$ . Justify your prediction in terms of periodic properties and Coulomb's law.

Chapter 10 Part 3:  $K_{sp}$ 

37. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

The majority of elements in the periodic table are metals. At 298 K most metals are solid and share similar physical properties.

(a) A student claims that gold is malleable because it is dense. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with wavelength 130 nm cause the ejection of 6  $s$  electrons from gold, but photons with wavelength 200 nm do not. A student claims this is because the photons with wavelength 200 nm are lower in energy than photons with wavelength 130 nm. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, X. The student takes a 100.0 g piece of metal X, heats it to 500.0 °C, then places it on a 1000.0 g block of ice at 0.0 °C. The ice partially melts, and the final temperature of the metal, ice, and melted water is 0.0 °C. The student calculates the experimental value of the specific heat capacity of metal X and records it as 0.90 J/(g · °C).

(c) Calculate the magnitude of the energy change of metal X during the experiment.

(d) Calculate the mass of ice melted by metal X. (The enthalpy of fusion of ice is 330 J/g.)

(e) A student claims that the energy transferred to the ice breaks the polar-covalent O — H bonds found between atoms within water molecules. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of  $\text{XF}_2$ , the fluoride salt of metal X, and mixes it with 100.0 mL of distilled water to form a saturated solution. The value of  $K_{sp}$  for  $\text{XF}_2$  is  $4.0 \times 10^{-9}$  at 298 K.

(f) Write the balanced chemical equation for the change that occurs when solid  $\text{XF}_2$  dissolves in water.

(g) Calculate the concentration of  $\text{F}^- (aq)$  ions in the saturated solution.

(h) Will the solubility of  $\text{XF}_2$  increase, decrease, or remain the same if it is dissolved in 100.0 mL of 0.10 M NaF instead of distilled water? Explain.

(i) The student gently heats 100.0 mL of the saturated solution of  $\text{XF}_2$ , evaporating enough water to reduce the volume of the solution to 75.0 mL. After the temperature of the solution returns to 298 K, will the concentration of  $\text{X}^{2+} (aq)$  ions in the solution be greater than, less than, or the same as when the volume of the solution was 100.0 mL? Justify your answer.

The first five ionization energies for metal X are shown in the following table.

Chapter 10 Part 3: K<sub>sp</sub>

| Ionization Energies of Metal X (kJ/mol) |        |       |        |        |
|-----------------------------------------|--------|-------|--------|--------|
| First                                   | Second | Third | Fourth | Fifth  |
| 736                                     | 1445   | 7730  | 10,500 | 13,600 |

(j) The ionic compound  $\text{XF}_3$  is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for  $\text{XCl}_2$  and for  $\text{XBr}_2$ .

| Dissociation Reaction                                             | Lattice Energy (kJ/mol) |
|-------------------------------------------------------------------|-------------------------|
| $\text{XCl}_2(s) \rightarrow \text{X}^{2+}(g) + 2 \text{Cl}^-(g)$ | 2524                    |
| $\text{XBr}_2(s) \rightarrow \text{X}^{2+}(g) + 2 \text{Br}^-(g)$ | 2440                    |

(k) Explain in terms of atomic properties and Coulomb's law why the lattice energy of  $\text{XCl}_2$  is greater than that of  $\text{XBr}_2$ .

Chapter 10 Part 3:  $K_{sp}$ 

38. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

The majority of elements in the periodic table are metals. At 298 K most metals are solid and share similar physical properties.

(a) A student claims that silver is a good conductor of electricity because the valence electrons are strongly attracted to the nucleus. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with frequency  $1.2 \times 10^{15}$  Hz cause the ejection of 5 s electrons from silver, but photons with frequency  $1.0 \times 10^{15}$  Hz do not. A student claims this is because the photons with frequency  $1.0 \times 10^{15}$  Hz are higher in energy than photons with frequency  $1.2 \times 10^{15}$  Hz. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, X. The student takes a 100.0 g piece of metal X, heats it to  $500.0^\circ\text{C}$ , then places it on a 1000.0 g block of ice at  $0.0^\circ\text{C}$ . The ice partially melts, and the final temperature of the metal, ice, and melted water is  $0.0^\circ\text{C}$ . The student calculates the experimental value of the specific heat capacity of metal X and records it as  $0.44 \text{ J}/(\text{g} \cdot ^\circ\text{C})$ .

(c) Calculate the magnitude of the energy change of metal X during the experiment.

(d) Calculate the mass of ice melted by metal X. (The enthalpy of fusion of ice is  $330 \text{ J/g}$ .)

(e) A student claims that the energy transferred to the ice overcomes attractions between  $\text{H}^+(\text{aq})$  ions and  $\text{OH}^-(\text{aq})$  ions in the ice. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of  $\text{Z}_2\text{SO}_4$ , a compound of a different metal, Z, and mixes it with 100.0 mL of water to form a saturated solution. The value of  $K_{sp}$  for  $\text{Z}_2\text{SO}_4$  is  $4.0 \times 10^{-6}$  at  $25^\circ\text{C}$ .

(f) Write the balanced chemical equation for the change that occurs when solid  $\text{Z}_2\text{SO}_4$  dissolves in water.

(g) Calculate the concentration of  $\text{SO}_4^{2-}(\text{aq})$  ions in the saturated solution.

(h) Will the solubility of  $\text{Z}_2\text{SO}_4$  increase, decrease, or remain the same if it is dissolved in 100.0 mL of  $0.10 \text{ M Na}_2\text{SO}_4$  instead of distilled water? Explain.

(i) The student then cools the saturated solution gently to a temperature of  $10^\circ\text{C}$ . If the enthalpy change of the dissolution process is negative, will the concentration of  $\text{Z}^+(\text{aq})$  ions in the solution be greater than, less than, or the same as when the temperature of the solution was  $25^\circ\text{C}$ ? Justify your answer.

The first five ionization energies for metal Z are shown in the following table.

Chapter 10 Part 3: K<sub>sp</sub>

| Ionization Energies of Metal Z (kJ/mol) |        |       |        |       |
|-----------------------------------------|--------|-------|--------|-------|
| First                                   | Second | Third | Fourth | Fifth |
| 403                                     | 2633   | 3787  | 5037   | 6603  |

(j) The ionic compound  $ZF_2$  is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for  $MCl_2$  and for  $MCl_3$ , which are chlorides of a different metal, M.

| Dissociation Reaction                          | Lattice Energy (kJ/mol) |
|------------------------------------------------|-------------------------|
| $MCl_2(s) \rightarrow M^{2+}(g) + 2 Cl^{-}(g)$ | 2631                    |
| $MCl_3(s) \rightarrow M^{3+}(g) + 3 Cl^{-}(g)$ | ?                       |

(k) A student predicts that the lattice energy of  $MCl_3$  is greater than the lattice energy of  $MCl_2$ . Do you agree with the student's prediction? Explain your answer in terms of atomic properties and Coulomb's law.

Chapter 10 Part 3:  $K_{sp}$ 

39. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

The majority of elements in the periodic table are metals. At 298 K most metals are solid and share similar physical properties.

(a) A student claims that silver is malleable because the valence electrons are strongly attracted to the nucleus. Do you agree or disagree with this claim? Justify your answer.

(b) In a photoelectron spectroscopy experiment, photons with frequency  $1.2 \times 10^{15}$  Hz cause the ejection of 5 s electrons from silver, but photons with frequency  $1.9 \times 10^{15}$  Hz cause the electrons to be ejected with greater energy. A student claims this is because the photons with frequency  $1.9 \times 10^{15}$  Hz are higher in energy than photons with frequency  $1.2 \times 10^{15}$  Hz. Do you agree or disagree with the student's claim? Justify your answer in terms of the energies of the photons.

A student investigates a pure metal, X. The student takes a 100.0 g piece of metal X, heats it to  $500.0^\circ\text{C}$ , then places it on a 1000.0 g block of ice at  $0.0^\circ\text{C}$ . The ice partially melts, and the final temperature of the metal, ice, and melted water is  $0.0^\circ\text{C}$ . The student calculates the experimental value of the specific heat capacity of metal X and records it as  $0.24 \text{ J}/(\text{g} \cdot ^\circ\text{C})$ .

(c) Calculate the magnitude of the energy change of metal X during the experiment.

(d) Calculate the mass of ice melted by metal X. (The enthalpy of fusion of ice is  $330 \text{ J/g}$ .)

(e) A student claims that the energy transferred to the ice overcomes only London dispersion forces between the water molecules in the solid ice. Do you agree or disagree with this claim? Justify your answer.

The student takes a sample of  $\text{Z}_2\text{S}$ , the sulfide of a different metal, Z, and mixes it with 100.0 mL of water to form a saturated solution. The value of  $K_{sp}$  for  $\text{Z}_2\text{S}$  is  $4.0 \times 10^{-30}$  at 298 K.

(f) Write the balanced chemical equation for the change that occurs when solid  $\text{Z}_2\text{S}$  dissolves in water.

(g) Calculate the concentration of  $\text{S}^{2-}(\text{aq})$  ions in the saturated solution.

(h) Will the solubility of  $\text{Z}_2\text{S}$  increase, decrease, or remain the same if it is dissolved in 100.0 mL of  $0.10 \text{ M Na}_2\text{S}$  instead of distilled water? Explain.

(i) The student then gently heats 100.0 mL of the saturated solution to a temperature of 315 K. If the enthalpy change of the dissolution process is negative, will the concentration of  $\text{Z}^+(\text{aq})$  ions in the solution be greater than, less than, or the same as when the temperature of the solution was 298 K? Justify your answer.

The first five ionization energies for metal Z are shown in the following table.



Chapter 10 Part 3: K<sub>sp</sub>

| Ionization Energies of Metal Z (kJ/mol) |        |       |        |       |
|-----------------------------------------|--------|-------|--------|-------|
| First                                   | Second | Third | Fourth | Fifth |
| 419                                     | 3052   | 4410  | 5900   | 8000  |

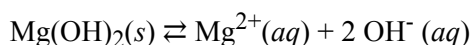
(j) The ionic compound  $ZF_2$  is not known to exist. Explain why, using the data in the table above to support your explanation.

For a compound, the energy required to separate the ions in a crystal lattice into individual gaseous ions is known as the lattice energy of the compound. The following table shows the dissociation reactions and the lattice energies for  $ZCl$  and for  $Z_2O$ .

| Dissociation Reaction                     | Lattice Energy (kJ/mol) |
|-------------------------------------------|-------------------------|
| $ZCl(s) \rightarrow Z^+(g) + Cl^-(g)$     | 829                     |
| $Z_2O(s) \rightarrow 2Z^+(g) + O^{2-}(g)$ | ?                       |

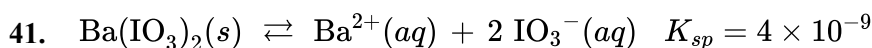
(k) A student predicts that the lattice energy of  $ZCl$  is greater than the lattice energy of  $Z_2O$ . If the ionic radii of  $Cl^-$  and  $O^{2-}$  are approximately the same, is the student's prediction correct? Explain your answer in terms of periodic properties and Coulomb's law.

40.



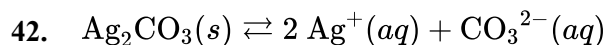
The exothermic dissolution of  $Mg(OH)_2(s)$  in water is represented by the equation above. The  $K_{sp}$  of  $Mg(OH)_2$  is  $1.8 \times 10^{-11}$ . Which of the following changes will increase the solubility of  $Mg(OH)_2$  in an aqueous solution?

- (A) Decreasing the pH
- (B) Increasing the pH
- (C) Adding  $NH_3$  to the solution
- (D) Adding  $Mg(NO_3)_2$  to the solution



According to the information about the dissolution of  $Ba(IO_3)_2(s)$  shown above, the correct value of  $S$ , the molar solubility of  $Ba(IO_3)_2(s)$ , can be calculated using with of the following mathematical relationships?

- (A)  $\frac{1}{2}S = 4 \times 10^{-9} M$
- (B)  $2S = 4 \times 10^{-9} M$
- (C)  $S^2 = 4 \times 10^{-9} M$
- (D)  $4S^3 = 4 \times 10^{-9} M$

Chapter 10 Part 3:  $K_{sp}$ 

The chemical equation above represents the equilibrium that exists in a saturated solution of  $\text{Ag}_2\text{CO}_3$ . If  $S$  represents the molar solubility of  $\text{Ag}_2\text{CO}_3$ , which of the following mathematical expressions shows how to calculate  $S$  based on  $K_{sp}$ ?

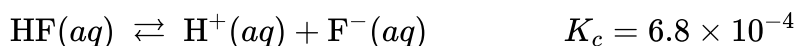
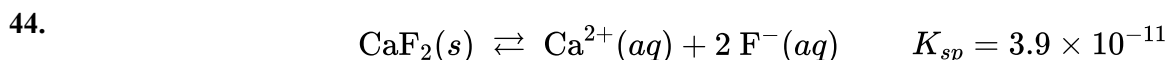
- (A)  $S = \sqrt{K_{sp}}$   
 (B)  $S = \sqrt{\frac{K_{sp}}{2}}$   
 (C)  $S = \sqrt[3]{\frac{K_{sp}}{2}}$   
 (D)  $S = \sqrt[3]{\frac{K_{sp}}{4}}$

43.

| Solution pH | $[\text{Ag}^+]_{eq}$  |
|-------------|-----------------------|
| 7.00        | $1.41 \times 10^{-4}$ |
| 8.00        | $1.41 \times 10^{-4}$ |
| 9.00        | $1.37 \times 10^{-4}$ |
| 10.00       | $1.00 \times 10^{-4}$ |
| 11.00       | $1.96 \times 10^{-5}$ |
| 12.00       | $2.00 \times 10^{-6}$ |
| 13.00       | $2.00 \times 10^{-7}$ |

A student investigates the effects of pH on the solubility of  $\text{AgOH}(s)$ , which dissolves in water according to the equation  $\text{AgOH}(s) \rightleftharpoons \text{Ag}^+(aq) + \text{OH}^-(aq)$ . The value for  $K_{sp}$  for  $\text{AgOH}$  is  $2.0 \times 10^{-8}$  at 298 K. The student places the same mass of  $\text{AgOH}(s)$  into 50.0 mL of different solutions with specific pH values and measures the concentration of  $\text{Ag}^+$  ions in each solution after equilibrium is reached. Based on the data in the table, what can be concluded about the solubility of  $\text{AgOH}$ ?

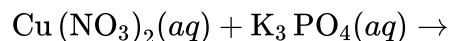
- (A) The solubility of  $\text{AgOH}$  is unaffected by pH because  $K_{sp}$  is a constant value at 298 K.  
 (B) The solubility of  $\text{AgOH}$  is unaffected by pH because it is the same value for pH 7.00 and pH 8.00.  
 (C)  $\text{AgOH}$  is more soluble at higher pH because lower concentrations of  $\text{OH}^-(aq)$  shift the solubility equilibrium toward  $\text{Ag}^+(aq)$  and  $\text{OH}^-(aq)$ .  
 (D)  $\text{AgOH}$  is less soluble at higher pH because higher concentrations of  $\text{OH}^-(aq)$  shift the solubility equilibrium toward solid  $\text{AgOH}$ .

Chapter 10 Part 3: K<sub>sp</sub>

The dissolution of calcium fluoride is represented by the equilibrium system above at 25 °C. The F<sup>−</sup> ion is produced when the weak acid HF dissociates. If solid calcium fluoride is added to equal volumes of the following solutions at 25 °C, in which solution will the most calcium fluoride dissolve?

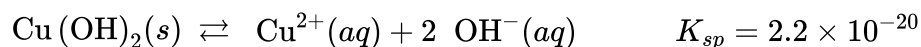
- (A) Pure distilled water
  - (B) 1 M HNO<sub>3</sub>(aq)
  - (C) 1 M NaOH(aq)
  - (D) A saturated aqueous CaF<sub>2</sub> solution
45. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is interested in studying the Cu<sup>2+</sup>(aq) ion, which is important in the environment.



(a) First, the student carries out a precipitation reaction. A solution of copper(II) nitrate is mixed with a solution of potassium phosphate, as represented above, and a blue precipitate forms. Write the balanced net ionic equation for the process.

(b) In a separate experiment the student decides to precipitate the Cu<sup>2+</sup>(aq) as Cu(OH)<sub>2</sub>(s). To make sure that Cu(OH)<sub>2</sub>(s) will precipitate, the student looks up the K<sub>sp</sub> value for Cu(OH)<sub>2</sub>(s), as shown below.



(i) Write the K<sub>sp</sub> expression for the dissolving of Cu(OH)<sub>2</sub>(s).

(ii) Calculate the concentration of Cu<sup>2+</sup>(aq) in a saturated Cu(OH)<sub>2</sub> solution in mol/L.

(c) The student incorrectly proposes that lowering the pH of the saturated Cu(OH)<sub>2</sub> solution will increase the amount of precipitate that forms. Explain why the student is incorrect.

Chapter 10 Part 3: K<sub>sp</sub>

46.

| Compound                 | $K_{sp}$ at 298 K  |
|--------------------------|--------------------|
| $\text{Ag}_2\text{SO}_4$ | $1 \times 10^{-5}$ |
| $\text{PbSO}_4$          | $1 \times 10^{-8}$ |

A 1.0 L solution  $\text{AgNO}_3(aq)$  of and  $\text{Pb}(\text{NO}_3)_2(aq)$  has a  $\text{Ag}^+$  concentration of 0.020 M and a  $\text{Pb}^{2+}$  concentration of 0.0010 M. A 0.0010 mol sample of  $\text{K}_2\text{SO}_4(s)$  is added to the solution. Based on the information in the table above, which of the following will occur? (Assume that the volume change of the solution is negligible.)

- (A) No precipitate will form.
- (B) Only  $\text{Ag}_2\text{SO}_4(s)$  will precipitate.
- (C) Only  $\text{PbSO}_4(s)$  will precipitate.
- (D) Both  $\text{Ag}_2\text{SO}_4(s)$  and  $\text{PbSO}_4(s)$  will precipitate.

47.

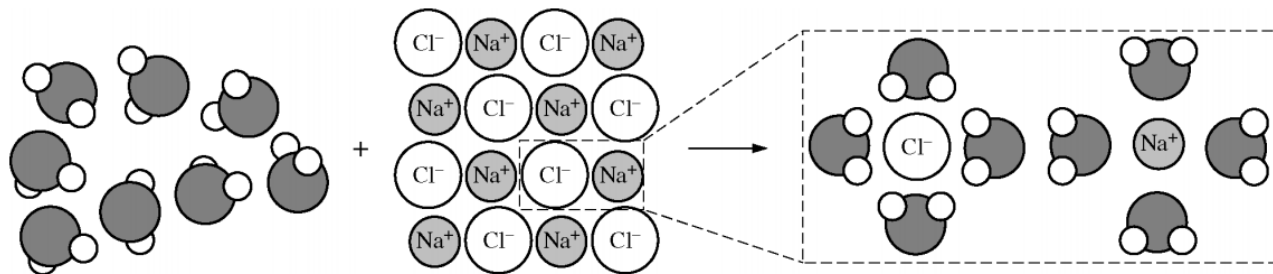
| Compound                 | $K_{sp}$              |
|--------------------------|-----------------------|
| $\text{Fe}(\text{OH})_2$ | $4.9 \times 10^{-17}$ |
| $\text{Cu}(\text{OH})_2$ | $1.6 \times 10^{-19}$ |
| $\text{Co}(\text{OH})_2$ | $1.1 \times 10^{-15}$ |

Which of the following ranks the compounds listed in the table above in order of increasing solubility?

- (A)  $\text{Fe}(\text{OH})_2 < \text{Cu}(\text{OH})_2 < \text{Co}(\text{OH})_2$
- (B)  $\text{Fe}(\text{OH})_2 < \text{Co}(\text{OH})_2 < \text{Cu}(\text{OH})_2$
- (C)  $\text{Co}(\text{OH})_2 < \text{Fe}(\text{OH})_2 < \text{Cu}(\text{OH})_2$
- (D)  $\text{Cu}(\text{OH})_2 < \text{Fe}(\text{OH})_2 < \text{Co}(\text{OH})_2$

Chapter 10 Part 3: K<sub>sp</sub>

48.



The process of dissolution of  $\text{NaCl}(s)$  in  $\text{H}_2\text{O}(l)$  is represented in the diagram above. Which of the following summarizes the signs of  $\Delta H^\circ$  and  $\Delta S^\circ$  for each part of the dissolution process?

| Breaking solvent-solvent interactions | Breaking solute-solute interactions | Forming solute-solvent interactions |
|---------------------------------------|-------------------------------------|-------------------------------------|
| $\Delta H^\circ$ $\Delta S^\circ$     | $\Delta H^\circ$ $\Delta S^\circ$   | $\Delta H^\circ$ $\Delta S^\circ$   |
| (A)      +      +                     | +      +                            | –      –                            |
| (B)      +      +                     | +      +                            | –      +                            |
| (C)      –      –                     | –      –                            | +      +                            |
| (D)      –      +                     | –      +                            | +      –                            |

49.

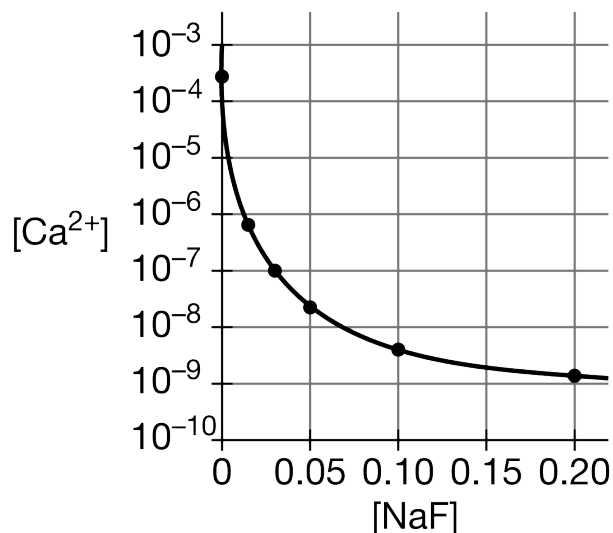
| Solution | Solute | $K_{sp}$ at $25^\circ\text{C}$ |
|----------|--------|--------------------------------|
| X        | AgBr   | $5.0 \times 10^{-13}$          |
| Y        | AgCl   | $1.8 \times 10^{-10}$          |
| Z        | AgI    | $8.3 \times 10^{-17}$          |

Three saturated solutions (X, Y, and Z) are prepared at  $25^\circ\text{C}$ . Based on the information in the table above, which of the following lists the solutions in order of increasing  $[\text{Ag}^+]$  ?

- (A)  $X < Z < Y$   
 (B)  $Y < X < Z$   
 (C)  $Z < Y < X$   
 (D)  $Z < X < Y$

Chapter 10 Part 3: K<sub>sp</sub>

50.



Which of the following statements best explains the trend in the molar solubility of  $\text{CaF}_2(s)$  in various concentrations of  $\text{NaF}(aq)$  shown in the graph above?

- (A) Higher concentrations of  $\text{F}^-(aq)$  allow less  $\text{CaF}_2(s)$  to dissolve before the solution becomes saturated.
- (B) Higher concentrations of  $\text{Na}^+(aq)$  lead to more favorable attractions between  $\text{Na}^+(aq)$  ions and  $\text{F}^-(aq)$  ions, thus less  $\text{CaF}_2(s)$  dissolves.
- (C) Higher concentrations of  $\text{F}^-(aq)$  produce more  $\text{OH}^-(aq)$  through hydrolysis, causing  $\text{Ca}(\text{OH})_2(s)$  to precipitate, thus more  $\text{CaF}_2(s)$  dissolves.
- (D) Higher concentrations of  $\text{F}^-(aq)$  cause the formation of the  $\text{CaF}_4^{2-}(aq)$  complex ion, which allows more  $\text{CaF}_2(s)$  to dissolve.

51.  $\text{CuBr}(s) \rightleftharpoons \text{Cu}^+(aq) + \text{Br}^-(aq)$

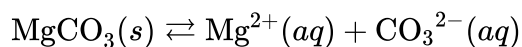
Shown above is the chemical equation for the dissolution of the slightly soluble salt  $\text{CuBr}(s)$ . Its  $K_{sp}$  value in pure water was experimentally determined.  $\text{CuBr}$  was found to be much less soluble in a  $0.001\text{ M NaBr}$  solution than in pure water. Which of the following correctly explains the decrease in solubility of  $\text{CuBr}$  in  $0.001\text{ M NaBr}$ ?

- (A) The concentration of water is much less in the  $\text{NaBr}$  solution than in pure water, reducing the rate of dissolution of the salt.
- (B) The presence of  $\text{Na}^+$  ions in the solution inhibits the dissolution of salts.
- (C) The presence of additional  $\text{Br}^-$  already in the solution decreases the value of the  $K_{sp}$  for  $\text{CuBr}$ , causing the solubility to decrease.
- (D) The presence of additional  $\text{Br}^-$  ions already in the solution means equilibrium will be reached when much less  $\text{CuBr}$  has dissolved.

Chapter 10 Part 3: K<sub>sp</sub>

52.

| Moles of HCl added   | Molar solubility of MgCO <sub>3</sub> |
|----------------------|---------------------------------------|
| 0                    | $2.6 \times 10^{-3}$                  |
| $2.7 \times 10^{-4}$ | $3.3 \times 10^{-3}$                  |
| $4.6 \times 10^{-4}$ | $5.5 \times 10^{-3}$                  |
| $6.0 \times 10^{-4}$ | $6.9 \times 10^{-3}$                  |



A saturated solution of MgCO<sub>3</sub> at equilibrium is represented by the equation above. Four different saturated solutions were prepared and kept at the same temperature. A given amount of HCl was added to each solution and data were collected to calculate the molar solubility of MgCO<sub>3</sub> as shown in the table above. Which of the following can be concluded from the data?

- (A) The molar solubility of MgCO<sub>3</sub> decreases with increasing acidity (lower pH).
- (B) The molar solubility of MgCO<sub>3</sub> increases with increasing acidity (lower pH).
- (C) The pH or acidity of the solution has a negligible effect on the molar solubility of MgCO<sub>3</sub>.
- (D) From the data it is not possible to determine the effect of changes in acidity on the molar solubility of MgCO<sub>3</sub>.

Chapter 10 Part 3: K<sub>sp</sub>

53. For parts of the free response question that require calculations, clearly show the method and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

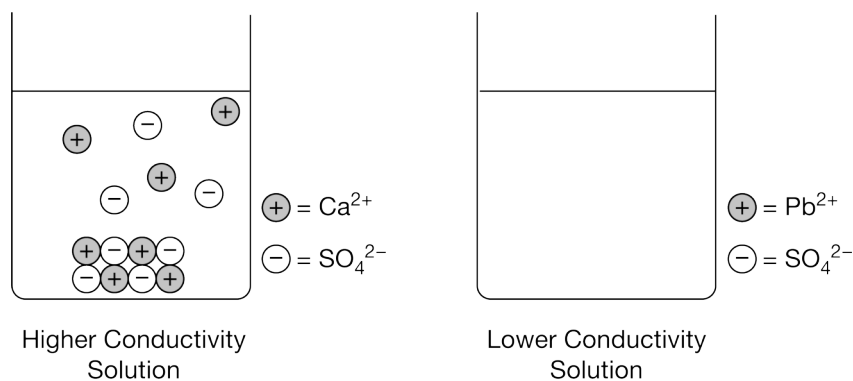
A student is studying the properties of  $\text{CaSO}_4$  and  $\text{PbSO}_4$ . The student has samples of both compounds, which are white powders.

(a) The student tests the electrical conductivity of each solid and observes that neither solid conducts electricity. Describe the structures of the solids that account for their inability to conduct electricity.

The student places excess  $\text{CaSO}_4(s)$  in a beaker containing 100 mL of water and places excess  $\text{PbSO}_4(s)$  in another beaker containing 100 mL of water. The student stirs the contents of the beakers and then measures the electrical conductivity of the solution in each beaker. The student observes that the conductivity of the solution in the beaker containing the  $\text{CaSO}_4(s)$  is higher than the conductivity of the solution in the beaker containing the  $\text{PbSO}_4(s)$ .

(b) Which compound is more soluble in water,  $\text{CaSO}_4(s)$  or  $\text{PbSO}_4(s)$ ? Justify your answer based on the results of the conductivity test.

The left side of the diagram below shows a particulate representation of the contents of the beaker containing the  $\text{CaSO}_4(s)$  from the solution conductivity experiment.



(c) Draw a particulate representation of  $\text{PbSO}_4(s)$  and the ions dissolved in the solution in the beaker on the right in the diagram. Draw the particles to look like those shown to the right of the beaker. Draw an appropriate number of dissolved ions relative to the number of dissolved ions in the beaker on the left.

(d) The student attempts to increase the solubility of  $\text{CaSO}_4(s)$  by adding 10.0 mL of 2 M  $\text{H}_2\text{SO}_4(aq)$  to the beaker, and observes that additional precipitate forms in the beaker. Explain this observation.